

Preprint

Afterglow Dark Energy: Apparent Phantom Crossing from Hidden Confinement, Tested Against Planck, DESI DR2, and Pantheon+

Thomas G. Martin ^{a,1,*}, In-Gyu Koh^{b,1}

^a*Independent Researcher, North Vancouver, BC, Canada*

^b*Independent Researcher, Daejeon, South Korea*

Abstract

DESI DR2 and Pantheon+ strengthen the case for a dynamic dark sector, yet in standard General Relativity such dynamics are often modeled by phantom crossings, finely tuned masses, or ad hoc interaction shells. We study a different starting point. If gravity is a macroscopic thermodynamic equation of state in Jacobson’s sense, a Lorentz-invariant equilibrium vacuum contributes no null heat flow and must be removed from the macroscopic source law: gravity responds to the non-equilibrium stress tensor $\tau_{\mu\nu}$ alone, and cosmic acceleration then requires a persistent out-of-equilibrium cosmic medium. We model that medium as the slowly relaxing thermodynamic exhaust of a hidden pure $SU(2)_D$ sector that confines at a dynamically generated meV scale. The gapped post-confinement phase is described causally by a memory variable Σ , giving an intrinsic non-phantom equation of state $w_X = -1 + 1/(3c_D)$. Matter loading renormalizes the relaxation rate through a fixed signed kernel $\Psi(r)$ that vanishes at matter–dark-energy equality, so the exchange law, its sign reversal at equality, and the apparent phantom crossing of the effective equation of state follow from the relaxation dynamics with a single amplitude β . We implement the theory as an independent memory species in the modified Boltzmann code `CLASS_SYMT`, and constrain it with Planck NPIPE CMB and lensing, DESI DR2 BAO, and Pantheon+

*Corresponding author

Email addresses: `thomas.glenn.martin@gmail.com` (Thomas G. Martin )
`ingyukoh2@gmail.com` (In-Gyu Koh)

¹Formerly: Department of Physics, Korea Advanced Institute of Science and Technology (KAIST).

across three converged MCMC branches. The isolated $\beta = 0$ branch survives the full stack; the interacting branch bounds the data-resolved coupling at $g \equiv \beta/c_D < 0.20$ at 95% credibility, a constraint set through geometry rather than through the growth channel in which g is defined; the microscopic amplitude the chains inform is the initial pressure deficit $\varepsilon_\Sigma = 0.012^{+0.023}_{-0.008}$; and Bayesian evidence relative to Λ CDM is inconclusive on all three branches, $|\Delta \log Z| < 1$, with a mild lean toward finite c_D . A pre-registered growth extension with redshift-space distortions and cosmic shear, forecast to roughly halve the coupling bound, is timed to the DESI 2025-2026 releases.

Keywords: dark energy theory, cosmological perturbation theory, cosmological parameters from CMBR

1. Introduction

The cosmological constant problem [1] becomes clearer when split into two separate questions:

- (i) Why do the large equilibrium vacuum contributions of quantum fields and phase transitions fail to curve spacetime catastrophically?
- (ii) Why does the late universe nevertheless contain an accelerating component with present scale $\rho_{X,0}^{1/4} \simeq 2.3 \text{ meV}$?

Those questions need not have the same answer. In the framework studied here, they do not.

The standard presentation of the problem assumes from the start that gravity is fundamentally an action-principle geometry sourced by the full stress tensor. But if gravity is instead a macroscopic thermodynamic equation of state, the source law must be re-read. In Jacobson’s horizon derivation, the relevant local quantity is the null heat flow across causal horizons, and a Lorentz-invariant equilibrium vacuum contributes exactly zero to that flow. On that reading, the vacuum catastrophe is not something to be hidden after the fact. It is a sign that the macroscopic source of gravity has been written incorrectly.

This changes the late-time dark-energy problem in a useful way. Once the equilibrium baseline is subtracted from the source term, the observed acceleration can no longer be attributed to an exactly equilibrium vacuum. It must come from a medium that is still out of equilibrium today. Within

that premise, a persistent cosmic non-equilibrium source is not optional. It is mandatory.

The model studied here supplies that source with three linked ingredients. First, the macroscopic source of gravity is the non-equilibrium stress tensor $\tau_{\mu\nu}$, not the equilibrium vacuum baseline. Second, a hidden pure $SU(2)_D$ sector generates a meV confinement scale dynamically through dimensional transmutation [2]. Third, after confinement the infrared dark sector is modeled as a gapped arrested medium whose long-wavelength state is characterized by a scalar pressure-deficit variable Σ . The exact-equilibrium remainder of that medium is subtracted into the equilibrium baseline, while the slowly relaxing non-equilibrium tail gravitates.

The isolated arrested phase already has a critical consequence:

$$w_X = -1 + \frac{1}{3c_D}, \quad (1)$$

with c_D the late-time Deborah number. That is enough to produce an accelerating sourced branch when $c_D > 1/2$. The open-system extension developed here is designed to preserve that homogeneous branch while allowing the surrounding dark matter environment to modify the *rate* at which the tail relaxes and the *sign* of the exchange across the matter-dominated and dark-energy-dominated eras.

A recent multi-model comparison using DESI DR2, Pantheon+, Planck, and ACT provides a useful qualitative rubric for what late-time dark-energy theories should at least be able to mimic: a phantom tendency at higher redshift, a quintessence tendency at lower redshift, and possible late-time transfer from dark energy to matter, while neither decisively beating Λ CDM in Bayesian evidence nor resolving the Hubble tension outright [3]. We use that pattern as the phenomenological context, and we confront the model with data of exactly that class: Section 5 reports three converged MCMC branches (Table 5) against Planck (NPIPE CamSpec TTTEEE, low- ℓ , and lensing), DESI DR2 BAO, and Pantheon+.

The DESI-era model literature has, in parallel, converged on three empirical lessons. Stable intrinsic phantom crossing is forbidden for a single field: at the crossing the sound speed vanishes and the phantom side develops gradient instabilities [4], and in single-field cosmologies with Hubble-modulated kinetic response, ghost freedom makes the divide an invariant, dynamically stable manifold of the flow [5]. Apparent crossing therefore requires internal degrees of freedom or a dark-sector coupling that an interaction-blind

observer misreads as super-acceleration [6–9]. And the couplings the data mildly prefer reverse sign at dark-sector equality [10]. The afterglow framework produces all three as consequences of one structure: the intrinsic equation of state is non-phantom by construction, the exchange produces the apparent crossing, and the sign reversal follows from the kernel’s zero at matter–dark-energy equality. The closest structural neighbors are the dissipative scenarios recently confronted with the same data, in which bulk-viscous or weakly dissipated dark energy reproduces the evolving equation of state and the apparent crossing [11, 12]. Those models postulate the dissipative function, a viscosity coefficient $\zeta(\rho)$ or a chosen friction on a quintessence field; here the relaxation structure, the kernel, and the c_D regime are fixed in advance by the hidden-sector construction, so the dissipative mechanism is derived rather than parametrized. The two families fork on an observable as well: field dissipation feeds the matter sector in one direction throughout, whereas the afterglow exchange reverses sign at equality, the structure that sign-reversing couplings and non-parametric reconstructions of the DESI data now probe directly [10, 13].

What is exact and what is assumed should therefore be separated carefully.

- Motivated by horizon thermodynamics: a Lorentz-invariant equilibrium vacuum contributes no null heat flow. The equilibrium-subtraction postulate (Postulate ES) elevates this observation to a source-law principle, removing the equilibrium baseline before gravity is sourced.
- Exact on the homogeneous post-transition arrested branch once $\tau_\Sigma = c_D/H$ is assumed: the confined phase splits as $\rho_{\text{conf}} = V_{D,\text{eq}} + 3c_D\Sigma$ and the sourced branch obeys (1).
- Constitutive but restricted: the open-system correction depends on one dimensionless amplitude β and one fixed sign-changing kernel $\Psi(r)$.
- Constitutive and tested numerically in Section 5: the late-time scaling $\tau_\Sigma = c_D/H$, the EFT activation of the post-confinement phase, the perturbative transfer frame, and the stabilised scalar closure with rest-frame sound speed $c_{s,X}^2$ (held at $c_{s,X}^2 = 1/2$ throughout Phase 3 and released in the growth extension).

The theory is written in a form explicit enough to code, test, and falsify, and it has now been coded, tested, and confronted with data. The model

uses one intrinsic constitutive parameter c_D , one interaction amplitude β , one fixed signed kernel $\Psi(r)$, and one independent scalar perturbation variable $\delta\Sigma$; it does not require a free response shell with multiple adjustable functions. Four results summarize the present status. First, the isolated $\beta = 0$ branch survives a full Planck+DESI DR2+Pantheon+ Boltzmann analysis in converged chains ($R - 1 \leq 6.7 \times 10^{-3}$; Table 5). Second, the non-crossing interacting branch bounds the data-resolved coupling at $g \equiv \beta/c_D < 0.20$ at 95% credibility, a constraint set through the geometry and CMB channels rather than through the growth channel in which g is defined. Third, the Bayesian evidence relative to Λ CDM is inconclusive on all three branches, $|\Delta \log Z| < 1$ on the Jeffreys scale, with a mild positive lean toward finite c_D consistent with the current mild preference for dark-energy dynamics. Fourth, the one microscopic amplitude the present chains inform, the initial pressure deficit of the arrested phase, is constrained to $\varepsilon_\Sigma = 0.012^{+0.023}_{-0.008}$, an order- 10^{-2} value consistent with a plausible hidden-sector transition. The growth test that would discriminate the interacting branch is pre-registered: the Phase 3.5 configuration (redshift-space distortions and cosmic shear, freed $c_{s,X}^2$, sampled transition inputs, priors fixed in Appendix F) is timed against the DESI 2025–2026 full-shape and lensing releases, with DR3 (expected 2027) the dated horizon for the $w_X = -1 + 1/(3c_D)$ prediction.

The paper is organized as follows. Section 2 redefines the source of gravity in thermodynamic language. Section 3 introduces the Yang-Mills trigger that makes a persistent non-equilibrium source plausible. Section 4 applies causal thermodynamics to the post-confinement medium and derives the interacting afterglow. Section 5 formulates the cosmological equations and linear perturbations, describes the `CLASS_SYMT` implementation and the MCMC methodology, and reports the Phase 3 constraints. Section 6 concludes. The staged bring-up sequence, the pre-run validation checklist and its verification status, recommended priors for the growth-extension chains, and the sample-by-sample microscopic reconstruction recipe are collected in Appendix F.

2. The source of gravity

2.1. Horizon thermodynamics and the null-projection observation

This section develops the starting point of the framework. It builds on Jacobson’s thermodynamic derivation of gravity [14], and the broader emergent-gravity perspective of Padmanabhan [15–17].

At any spacetime point one may consider the local causal horizon associated with an accelerated observer. In Jacobson's 1995 derivation, the Einstein equation is recovered from the Clausius relation $\delta Q = T dS$ applied to pencils of null generators threading those horizons. The heat flow entering the derivation is controlled by the null projection

$$T_{\mu\nu} k^\mu k^\nu, \quad (2)$$

where k^μ is tangent to the null generators. A Lorentz-invariant equilibrium vacuum has the form

$$T_{\mu\nu}^{\text{eq}} = -\rho_{\text{eq}} g_{\mu\nu}. \quad (3)$$

Contracting with any null vector immediately gives

$$T_{\mu\nu}^{\text{eq}} k^\mu k^\nu = -\rho_{\text{eq}} g_{\mu\nu} k^\mu k^\nu = 0. \quad (4)$$

Equation (4) is exact and uncontroversial: the equilibrium vacuum is invisible to the null heat flow that enters Jacobson's derivation. In standard general relativity this fact has no structural consequence, because the Einstein equation is obtained from an action principle in which the full $T_{\mu\nu}$ appears as the source by construction. The null-projection zero is then an accidental kinematic property of one sector.

The present framework takes a different view. If gravity *is* a macroscopic thermodynamic equation of state — rather than a fundamental action-principle geometry that happens to admit a thermodynamic re-reading — then the source law should be written in terms of the quantities that actually enter the thermodynamic construction. Because the equilibrium vacuum contributes nothing to the null heat flow that *defines* gravitational thermodynamics, it should not appear on the right-hand side.

We elevate this interpretive step to an explicit postulate.

Postulate (Equilibrium-Subtraction, ES).. The macroscopic source of gravity is the non-equilibrium stress tensor $\tau_{\mu\nu}$, defined by splitting the total stress-energy as

$$T_{\mu\nu} = T_{\mu\nu}^{\text{eq}} + \tau_{\mu\nu}, \quad \tau_{\mu\nu} := T_{\mu\nu} - T_{\mu\nu}^{\text{eq}}, \quad (5)$$

where $T_{\mu\nu}^{\text{eq}}$ is the total Lorentz-invariant equilibrium vacuum contribution. Only $\tau_{\mu\nu}$ enters the macroscopic field equation.

Two clarifications are essential.

First, Postulate ES does *not* follow from the null-projection identity (4) alone. Jacobson’s original derivation recovers the standard Einstein equation precisely because the full $T_{\mu\nu}$ is identified with the thermodynamic heat source. Restricting the source to $\tau_{\mu\nu}$ requires the additional physical commitment that gravity is *exclusively* a macroscopic equation of state of the kind that the null heat flow defines. The null-projection zero motivates Postulate ES; it does not entail it. The postulate is therefore an axiom of this framework, not a theorem.

Second, only a separately conserved, time-independent equilibrium sector is subtracted. Dynamical fields that are rolling, dissipating, or exchanging energy remain part of $\tau_{\mu\nu}$. In particular, inflation driven by a rolling scalar is not erased by Postulate ES; only its strictly constant equilibrium baseline would drop out.

2.2. The modified field equation and branch selection

Given Postulate ES, the macroscopic field equation takes the form

$$G_{\mu\nu} + \Lambda_{\text{ref}} g_{\mu\nu} = \frac{8\pi G}{c^4} \tau_{\mu\nu}. \quad (6)$$

The constant Λ_{ref} labels the curvature of the equilibrium branch — the geometry of the universe in the limit where all non-equilibrium sources are switched off. In principle its value is set by the equilibrium configuration and need not vanish. In this paper we work on the flat branch,

$$\Lambda_{\text{ref}} = 0, \quad G_{\mu\nu} = \frac{8\pi G}{c^4} \tau_{\mu\nu}. \quad (7)$$

The flat-branch choice is not derived here. We adopt it as the simplest consistent starting point: the equilibrium vacuum neither curves spacetime nor accelerates the expansion. A more fundamental selection principle for Λ_{ref} — whether from the structure of the equilibrium state itself, from a landscape measure, or from boundary conditions — is an open question that we set aside. What matters for the late-time dynamics is that Λ_{ref} is a fixed integration constant, not a dynamical source. The entire subsequent construction depends only on the structure of $\tau_{\mu\nu}$.

Equation (7) is not merely an algebraic reshuffling of a cosmological constant, and the construction is a thermodynamic relative of the trace-free and unimodular reformulations in which the vacuum is removed from the source

law at the level of the field equations [18]. The physical content of Postulate ES is that the equilibrium baseline is removed *before* the macroscopic source law is written, not subtracted after the fact.

2.3. The mandate for a persistent non-equilibrium source

Equation (7) reorganizes the late-time problem in a critical way. If the observed acceleration is not sourced by the equilibrium vacuum, then the universe must contain some medium whose non-equilibrium stress remains cosmologically relevant today. A successful dark-energy model in this framework cannot be a clever way of hiding the vacuum. It must be a concrete mechanism for sustaining a long-lived $\tau_{\mu\nu}$.

That observation sets the target for the rest of the paper. The source must satisfy three conditions. It must be near vacuum so that it can drive acceleration. It must be genuinely non-equilibrium so that it belongs on the right-hand side of Eq. (7). And it must relax slowly enough to survive for billions of years. Ordinary Standard Model plasmas fail the last requirement. We therefore look to a hidden confining sector whose post-transition medium is gapped and thermodynamically arrested.

3. Sourcing the friction: a hidden Yang-Mills sector

3.1. Why the source cannot be ordinary matter

The mandate from Section 2 is already restrictive. The source of today's acceleration must be a cosmic medium that is persistently out of equilibrium, yet close enough to vacuum to drive accelerated expansion. Ordinary visible-sector plasmas are the wrong objects: they thermalize too quickly, their late-time equation of state is not vacuum-like, and they do not naturally carry a billion-year relaxation time. What is needed instead is a medium that experienced a large thermodynamic shock, then entered a phase with a gapped spectrum and suppressed equilibration.

A confining hidden gauge theory supplies precisely that logic; hidden non-Abelian sectors of this type, and the gapped glueball spectra they confine into, are a well-studied family [19–22]. Above confinement it behaves as a relativistic dark-radiation bath. Near confinement it undergoes a strongly non-conformal transition. Below confinement its infrared degrees of freedom are gapped, so the medium can retain memory of the transition and relax only slowly. In that setting, a persistent dark-sector contribution to $\tau_{\mu\nu}$ is

not a guess added for convenience. It is the most economical way to satisfy the mandate created by Eq. (7).

Status of the Yang-Mills origin.. The hidden gauge-theory sector provides a concrete physical motivation for a persistent non-equilibrium source, but at the level of the present data stack it is not a tested component of the model. The late-time EFT that the MCMC chains constrain — a fluid with $w_X = -1 + 1/(3c_D)$, an energy exchange $Q = -g H \Psi(r) \rho_X$, and an independent memory perturbation $s = \delta\Sigma$ — does not require a Yang-Mills Lagrangian. The dimensional-transmutation story becomes testable only when the microscopic parameters $(\Delta N_{\text{eff}}, z_c, x_c)$ are sampled directly against data that resolve the transition epoch. Until then, the Yang-Mills sector should be read as the theoretical motivation for why a suitable source might exist, not as a data-validated claim.

3.2. Microscopic dark sector

The hidden sector is taken to be pure Yang-Mills with no light matter,

$$\mathcal{L}_D = -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu}, \quad a = 1, 2, 3. \quad (8)$$

Its one-loop beta function is

$$\mu \frac{dg_D}{d\mu} = -\frac{11N}{48\pi^2} g_D^3. \quad (9)$$

For pure $\text{SU}(2)_D$,

$$b_0 = \frac{11N}{3} = \frac{22}{3}, \quad (10)$$

and integrating the renormalization-group equation gives

$$\Lambda_D = \mu \exp\left[-\frac{8\pi^2}{b_0 g_D^2(\mu)}\right] = \mu \exp\left[-\frac{2\pi}{b_0 \alpha_D(\mu)}\right]. \quad (11)$$

With $\mu = M_{\text{Pl}} = 2.435 \times 10^{18} \text{ GeV}$ and $\Lambda_D \sim 2.3 \text{ meV}$ one finds

$$\alpha_D(M_{\text{Pl}}) \simeq \frac{2\pi}{b_0 \ln(M_{\text{Pl}}/\Lambda_D)} \simeq 0.0124. \quad (12)$$

Thus the required meV scale can arise dynamically through dimensional transmutation rather than by fine-tuning a rigid microscopic mass parameter.

3.3. Pre-confinement thermal history

Let the dark and visible sectors have temperature ratio

$$\xi \equiv \frac{T_D}{T_\gamma}. \quad (13)$$

Above confinement, the hidden sector is a bath of dark gluons with

$$g_D = 2(N^2 - 1) = 6 \quad (14)$$

relativistic bosonic degrees of freedom. Its contribution to the effective number of relativistic species is

$$\Delta N_{\text{eff}} = \frac{4}{7} g_D \left(\frac{T_D}{T_\nu} \right)^4 = \frac{4}{7} g_D \left(\frac{11}{4} \right)^{4/3} \xi^4 = 13.2 \xi^4. \quad (15)$$

The confinement transition occurs when

$$T_{D,c} = x_c \Lambda_D, \quad x_c = \mathcal{O}(1), \quad (16)$$

so the visible-sector redshift of confinement is

$$1 + z_c = \frac{T_{\gamma,c}}{T_{\gamma,0}} = \frac{x_c \Lambda_D}{\xi T_{\gamma,0}} \simeq 9.8 x_c \left(\frac{\Lambda_D}{2.3 \text{ meV}} \right) \xi^{-1}. \quad (17)$$

Equations (15) and (17) form the basic bridge between the microscopic hidden sector and the late-time effective description.

4. The arrested medium

4.1. Memory variable, equilibrium matching, and isolated branch

Below confinement we model the coarse-grained confined phase by an isotropic stress tensor

$$T_{\text{conf}}^{\mu\nu} = \rho_{\text{conf}} u^\mu u^\nu + p_{\text{conf}} \Delta^{\mu\nu}, \quad \Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu, \quad (18)$$

with constitutive relation

$$p_{\text{conf}} = -\rho_{\text{conf}} + \Sigma. \quad (19)$$

The scalar $\Sigma \geq 0$ measures the pressure deficit away from exact equilibrium. Here it is not introduced as a convenient parametrization. It is the causal

memory variable required by Israel-Stewart-type thermodynamics once the post-confinement medium is gapped and cannot equilibrate instantaneously [23–25]. We work throughout in the Müller–Israel–Stewart causal framework rather than first-order Eckart theory, whose equilibria are generically unstable and whose high-frequency modes are superluminal [26]; the finite relaxation time carried by Σ is the ingredient that repairs both pathologies and keeps the coupled system causal and well posed [24].

In the isolated arrested regime the order parameter free-relaxes with finite relaxation time,

$$u^\mu \nabla_\mu \Sigma = -\frac{\Sigma}{\tau_\Sigma} = -\frac{\Theta}{3c_D} \Sigma, \quad \Theta \equiv \nabla_\mu u^\mu, \quad (20)$$

where the minimal late-time constitutive choice is

$$\tau_\Sigma = \frac{c_D}{H}, \quad c_D > 0. \quad (21)$$

In homogeneous FRW, where $\Theta = 3H$, Eq. (20) becomes

$$\dot{\Sigma} = -\frac{H}{c_D} \Sigma. \quad (22)$$

If the confined phase is isolated after the transition, local conservation gives

$$\dot{\rho}_{\text{conf}} + 3H(\rho_{\text{conf}} + p_{\text{conf}}) = 0 \quad \implies \quad \dot{\rho}_{\text{conf}} = -3H\Sigma. \quad (23)$$

Combining (22) and (23) yields

$$\frac{d\rho_{\text{conf}}}{d\Sigma} = 3c_D, \quad (24)$$

and therefore

$$\rho_{\text{conf}} = \rho_* + 3c_D \Sigma. \quad (25)$$

The integration constant is not left uninterpreted. In exact equilibrium $\Sigma \rightarrow 0$, Eq. (19) gives $p_{\text{conf}} \rightarrow -\rho_{\text{conf}}$, while Eq. (23) implies $\dot{\rho}_{\text{conf}} \rightarrow 0$. The residual piece is therefore vacuum-like,

$$T_{\text{eq}}^{\mu\nu} = -V_{D,\text{eq}} g^{\mu\nu}, \quad \delta V_{D,\text{eq}} = 0, \quad (26)$$

and we identify

$$\rho_* \equiv V_{D,\text{eq}}. \quad (27)$$

The sourced non-equilibrium tail is then defined by the equilibrium-subtracted stress tensor

$$\tau_X^{\mu\nu} := T_{\text{conf}}^{\mu\nu} - T_{\text{eq}}^{\mu\nu}. \quad (28)$$

The homogeneous post-transition branch is therefore

$$\rho_{\text{conf}} = V_{D,\text{eq}} + 3c_D \Sigma, \quad (29)$$

with sourced variables

$$\rho_X \equiv \rho_{\text{conf}} - V_{D,\text{eq}} = 3c_D \Sigma, \quad p_X \equiv p_{\text{conf}} + V_{D,\text{eq}} = -(3c_D - 1)\Sigma. \quad (30)$$

Its intrinsic equation of state is

$$w_X \equiv \frac{p_X}{\rho_X} = -1 + \frac{1}{3c_D}. \quad (31)$$

Thus the sourced tail is automatically non-phantom for every $c_D > 0$, and it accelerates whenever $c_D > 1/2$.

4.2. Derivation of the signed loading-unloading kernel

The post-confinement medium is not relaxing in a void. It shares the universe with cold dark matter, so its relaxation rate must feel an environmental load. At the homogeneous level the only natural low-derivative scalar for that load is the density ratio

$$r \equiv \frac{\rho_c}{\rho_X}. \quad (32)$$

The environmental correction to the relaxation rate should be a dimensionless function $\Psi(r)$ that encodes whether the dark sector is matter-loaded or dark-energy-dominated. We impose five requirements on Ψ .

1. *Dimensionless and bounded.* The correction enters multiplicatively in the relaxation rate and must not introduce a new scale.
2. *Asymptotic silence.* $\Psi(r) \rightarrow 0$ as $r \rightarrow 0$ and as $r \rightarrow \infty$, so that the isolated tail is recovered when either component becomes negligible.
3. *Sign change at dark-sector balance.* Ψ changes sign at $r = 1$.
4. *Loading-unloading convention.* $\Psi(r) > 0$ for $r > 1$ (matter-loaded regime) and $\Psi(r) < 0$ for $0 < r < 1$ (dark-energy-dominated regime).

5. *Lowest-degree polynomial.* Among all smooth functions satisfying requirements 1–4, we select the polynomial of minimal degree.

Write

$$x \equiv \frac{r}{1+r} \in [0, 1]. \quad (33)$$

Then $x = 0$ and $x = 1$ correspond to $r = 0$ and $r = \infty$, while $x = 1/2$ corresponds to $r = 1$. Requirements 1–4 force Ψ to vanish at $x = 0$, $x = 1/2$, and $x = 1$, and to be odd about $x = 1/2$. The lowest-degree polynomial with those three roots and that symmetry is

$$\Psi(x) = 4x(1-x)(2x-1). \quad (34)$$

Rewriting in terms of r gives

$$\Psi(r) = \frac{4r(r-1)}{(1+r)^3}. \quad (35)$$

It obeys

$$\Psi(r) > 0 \quad (r > 1), \quad \Psi(r) < 0 \quad (0 < r < 1), \quad \Psi(1) = 0, \quad (36)$$

and vanishes in both asymptotic regimes.

Requirement 5 is a constitutive choice, not a uniqueness theorem. Any function of the form $\Psi_n(x) = A_n x^n(1-x)^n(2x-1)$ with $n \geq 1$ satisfies requirements 1–4, as do non-polynomial alternatives such as $\sin(\pi x)(2x-1)$. The choice $n = 1$ is the minimal polynomial ansatz: it introduces no additional shape freedom beyond what the sign and boundary conditions already dictate, and it keeps the number of extrema (one positive maximum, one negative minimum) as small as possible. Adding higher-order terms would introduce additional free coefficients without a physical principle to fix them. In this sense, the kernel is selected by the same reasoning that favors the lowest-order constitutive law in any effective description: start with the simplest structure that captures the required qualitative behavior and let the data decide whether refinement is needed. The adequacy of the $n = 1$ choice is ultimately tested by the MCMC chains; should the data require a broader or narrower kernel profile, the natural extension is to allow n or a shape parameter to be sampled.

4.3. Mandatory open-system loading and derived exchange law

The open-system postulate of the theory is that matter loading renormalizes the relaxation rate of the same memory variable Σ :

$$u^\mu \nabla_\mu \Sigma = -\frac{\Theta}{3c_D} \left[1 - \beta \Psi(r) \right] \Sigma, \quad \beta \geq 0. \quad (37)$$

In homogeneous FRW this becomes

$$\dot{\Sigma} = -\frac{H}{c_D} \left[1 - \beta \Psi(r) \right] \Sigma. \quad (38)$$

The parameter β measures the environmental loading-unloading correction relative to the isolated relaxation rate. The branch $\beta = 0$ returns exactly to the isolated sourced tail.

Now treat the sourced tail as an open subsystem,

$$\nabla_\mu \tau_X^{\mu\nu} = -Q^\nu, \quad \nabla_\mu T_c^{\mu\nu} = Q^\nu. \quad (39)$$

The sign convention is: positive Q means energy flows from dark energy to matter (ρ_c increases, ρ_X decreases). In homogeneous FRW only the scalar Q enters, so

$$\dot{\rho}_X + 3H(\rho_X + p_X) = -Q \quad \implies \quad \dot{\rho}_X + \frac{H}{c_D} \rho_X = -Q. \quad (40)$$

Combining (38), (40), and the *imposed requirement* that the homogeneous branch relation (29) continue to hold in the open system gives

$$\frac{d\rho_{\text{conf}}}{d\Sigma} = \frac{-3H\Sigma - Q}{-\frac{H}{c_D}(1 - \beta\Psi)\Sigma} = 3c_D. \quad (41)$$

Solving (41) for Q yields the background exchange law

$$Q = -3\beta H \Psi(r) \Sigma = -\frac{\beta}{c_D} H \Psi(r) \rho_X. \quad (42)$$

This is the key structural point: once the homogeneous branch relation is preserved, the exchange law is not an independent ansatz. It is determined by the loading-unloading relaxation law. In the classification of coupled dark-sector models by Pourtsidou, Skordis, and Copeland [27], the after-glow interaction is an *energy-exchange* coupling — the transfer Q enters the

background continuity equations (44)–(45) — rather than a pure momentum-exchange (Type-3) coupling that leaves the background unchanged. What distinguishes it from a generic energy-exchange model is that Q is not a free function: it is fixed by the relaxation law up to the single amplitude β , and its sign is set by the kernel $\Psi(r)$ rather than chosen.

Physical coupling.. The combination

$$g \equiv \frac{\beta}{c_D} \quad (43)$$

is the only place β enters the homogeneous dynamics (42)–(45). In data analyses against likelihoods that probe c_D through its β -free terms (the intrinsic equation of state $w_X = -1 + 1/(3c_D)$ in the geometry channel, and the $1/c_D$ prefactor on the memory-mode evolution in the perturbation channel), the data-resolved coupling is g , not β separately. We retain β when describing the dynamics and report g when stating the data constraint; the two are the same statement in different parameter bases.

The background evolution equations are therefore

$$\dot{\rho}_X = -\frac{H}{c_D} \left[1 - \beta \Psi(r) \right] \rho_X, \quad (44)$$

$$\dot{\rho}_c + 3H\rho_c = -\frac{\beta}{c_D} H \Psi(r) \rho_X. \quad (45)$$

Hence the sourced relation $\rho_X = 3c_D \Sigma$ remains exact at the homogeneous background level in the interacting arrested regime.

Place in the interacting dark-energy taxonomy.. The exchange law (42) makes this an *energy-exchange* coupled dark-energy–dark-matter model: the background transfer $Q = -(\beta/c_D)H\Psi(r)\rho_X$ is non-vanishing and of the phenomenological $Q \propto H\rho$ family widely studied in the interacting-DE literature [28, 29]. In the Lagrangian-level classification of Pourtsidou, Skordis and Copeland [27], couplings divide into energy-exchange types (their Types 1 and 2, where the dark sectors swap energy density at the background level) and pure-momentum-exchange models (Type 3, e.g. dark scattering [30], in which the background densities evolve independently and the coupling acts only on the velocity/Euler sector). Because $Q \neq 0$ in the continuity equations (44)–(45), the present model sits in the energy-exchange

branch (Type 1/2), *not* in the pure-momentum Type 3 class. Its distinguishing feature within that branch is that the transfer is not a free-sign phenomenological ansatz but is fixed by the loading–unloading relaxation law through the signed kernel $\Psi(r)$: Q changes sign as $r = \rho_c/\rho_X$ crosses the extrema of Ψ , so the dark sector alternately loads from and unloads to the matter sector rather than transferring monotonically in one direction. This sign-changing, thermodynamically-derived Q is the structural element that separates the afterglow model from the constant-sign $Q \propto H\rho$ constructions and from momentum-only couplings.

4.4. *Intrinsic and effective equations of state, and the sign of transfer*

The physical stress tensor of the sourced tail is fixed by Eq. (31). Observers, however, often fit background data with an uncoupled dark-energy fluid defined by

$$\dot{\rho}_X + 3H(1 + w_{\text{eff}})\rho_X = 0. \quad (46)$$

Comparing (44) and (46) gives

$$w_{\text{eff}} = -1 + \frac{1 - \beta\Psi(r)}{3c_D}. \quad (47)$$

The intrinsic and effective equations of state therefore differ by

$$w_{\text{eff}} - w_X = -\frac{\beta}{3c_D}\Psi(r). \quad (48)$$

Three immediate consequences follow.

1. In the matter-loaded regime $r > 1$, one has $\Psi(r) > 0$, so $w_{\text{eff}} < w_X$. If $\beta\Psi(r) > 1$, then $w_{\text{eff}} < -1$ even though the physical branch remains non-phantom.
2. At dark-sector balance $r = 1$, one has $\Psi(1) = 0$, so $w_{\text{eff}} = w_X$ and $Q = 0$.
3. In the dark-energy-dominated regime $0 < r < 1$, one has $\Psi(r) < 0$, so $w_{\text{eff}} > w_X > -1$ and

$$Q > 0, \quad (49)$$

meaning that energy flows from dark energy to matter.

This is the qualitative sign pattern targeted by the model: effective phantom at higher redshift, quintessence-like behavior at lower redshift, and late-time dark-energy-to-matter transfer, all with a non-phantom intrinsic branch.

It is also useful to record the evolution equation for r in $N \equiv \ln a$:

$$\frac{dr}{dN} = r \left(-3 + \frac{1}{c_D} \right) - \frac{\beta}{c_D} \Psi(r)(1+r). \quad (50)$$

This equation makes the sign structure transparent. For $r > 1$, the interaction term pushes r downward faster. For $r < 1$, it pushes r upward. The late-time branch is therefore not automatic; its stability must be checked.

Because the kernel has a bounded maximum,

$$\Psi_{\max} = \frac{2\sqrt{3}}{9}, \quad (51)$$

one also has

$$w_{\text{eff}}^{\min} = -1 + \frac{1 - \beta\Psi_{\max}}{3c_D}. \quad (52)$$

Within the baseline crossing strip discussed in Section 5 and for the near-vacuum values $c_D \gtrsim 5$ favored for a late arrested phase, the effective phantom excursion is therefore expected to be mild, typically at the level of a few times 10^{-3} to 10^{-2} below -1 , not a deep phantom regime.

4.5. Asymptotics and the non-phantom future

Because the kernel vanishes in both asymptotic regimes,

$$\Psi(r) \rightarrow 0 \quad \text{for} \quad r \rightarrow \infty \quad \text{or} \quad r \rightarrow 0, \quad (53)$$

the isolated tail is recovered both early and late. The far-future branch relevant to this paper is the dark-energy-dominated one, $r \rightarrow 0$. Expanding Eq. (50) for small r gives

$$\Psi(r) = -4r + \mathcal{O}(r^2), \quad \frac{dr}{dN} = r \left[-3 + \frac{1+4\beta}{c_D} \right] + \mathcal{O}(r^2). \quad (54)$$

Therefore the decaying late-time branch $r \rightarrow 0$ is an attractor provided

$$c_D > \frac{1+4\beta}{3}. \quad (55)$$

This condition is mild. It is automatically satisfied throughout the near-vacuum regime $c_D \gtrsim 5$ even on the signed-kernel crossing branch.

Under Eq. (55), the late-time asymptotics reduce to the isolated tail,

$$\rho_X \propto a^{-1/c_D}, \quad w_{\text{eff}} \rightarrow w_X = -1 + \frac{1}{3c_D}. \quad (56)$$

If the sourced tail dominates the late universe, the Friedmann equation gives

$$a(t) \propto t^{2c_D}, \quad \rho_X(t) \propto t^{-2}. \quad (57)$$

The late future is therefore a slowly relaxing power-law acceleration sourced by the non-equilibrium tail itself, not a de Sitter state produced by an equilibrium vacuum. The model allows a transient effective $w_{\text{eff}} < -1$ episode, but not a true asymptotically phantom attractor.

4.6. Notation summary

Table 1 collects the symbols of the arrested-medium dynamics introduced in this section, with the equation where each is defined. The two free parameters of the homogeneous theory are the intrinsic relaxation index c_D and the interaction amplitude β ; everything else is either fixed (the kernel Ψ) or derived. The data-resolved combination $g \equiv \beta/c_D$ used when reporting constraints is defined in Eq. (43) and applied in Sec. 5.5.

5. Cosmological implementation and Phase 3 constraints

The equations in this section track the evolution of the non-equilibrium stress tensor $\tau_{\mu\nu}$ as it interacts with the cold-dark-matter environment. At background level this means ρ_X and p_X are the dark-sector contribution to $\tau_{\mu\nu}$. At perturbative level it means the memory variable must be evolved explicitly rather than collapsed into an instantaneous fluid closure.

5.1. Background FRW equations and transition activation

On the flat branch of Eq. (7), the local cosmological equations take the standard FRW form,

$$3M_{\text{Pl}}^2 H^2 = \rho_b + \rho_c + \rho_r + \rho_X + \rho_{dr,D}, \quad (58)$$

$$-2M_{\text{Pl}}^2 \dot{H} = \rho_b + \rho_c + \frac{4}{3}\rho_r + \frac{4}{3}\rho_{dr,D} + \rho_X + p_X, \quad (59)$$

Symbol	Meaning	Defined
Σ	Causal memory variable; pressure deficit from equilibrium, $\Sigma \geq 0$	Eq. (19)
τ_Σ	Relaxation time of the memory variable, $\tau_\Sigma = c_D/H$	Eq. (21)
c_D	Late-time Deborah number; dimensionless relaxation index, $c_D > 0$	Eq. (21)
Θ	Expansion scalar $\nabla_\mu u^\mu$ ($= 3H$ in FRW)	Eq. (20)
$\rho_* = V_{D,\text{eq}}$	Equilibrium (vacuum-like) energy subtracted from the source	Eq. (27)
$\tau_X^{\mu\nu}$	Equilibrium-subtracted non-equilibrium stress tensor	Eq. (28)
ρ_X, p_X	Sourced density and pressure, $\rho_X = 3c_D\Sigma$, $p_X = -(3c_D - 1)\Sigma$	Eq. (30)
w_X	Intrinsic equation of state, $w_X = -1 + 1/(3c_D)$	Eq. (31)
r	Dark-sector density ratio ρ_c/ρ_X	Eq. (32)
x	Compactified ratio $r/(1+r) \in [0, 1]$	Eq. (33)
$\Psi(r)$	Lowest-degree signed loading-unloading kernel, $4r(r-1)/(1+r)^3$	Eq. (34)
β	Dark-sector interaction amplitude (coupling strength)	Eq. (37)
Q	Homogeneous energy-exchange rate, $Q = -(\beta/c_D)H\Psi(r)\rho_X$	Eq. (42)
w_{eff}	Effective (uncoupled-fit) equation of state, $-1 + [1 - \beta\Psi]/(3c_D)$	Eq. (47)
N	Number of e -folds, $N = \ln a$	Eq. (50)

Table 1: Symbols of the arrested-medium dynamics (Sec. 4). The microscopic-reconstruction symbols ($\xi, \Lambda_D, \alpha_D, x_c, \varepsilon_\Sigma$) are defined where they first appear in Sec. 5.3ff., and the perturbation variables ($\delta\Sigma = s, \delta\rho_X, \nu_X, \chi, c_{s,X}^2$) in Sec. 5.3.

where each term on the right-hand side belongs to the active non-equilibrium source and $\rho_{dr,D}$ denotes the pre-confinement dark-radiation component. The sourced tail satisfies

$$\rho_X + p_X = \Sigma = \frac{\rho_X}{3c_D}. \quad (60)$$

The novel background equations are therefore Eqs. (44), (45), and (35).

The arrested-phase effective theory begins after confinement. What we model here is the activation of the *post-confinement effective theory*, not a first-principles microphysical description of the confinement transition itself. It is therefore helpful to separate the bookkeeping from the constitutive late-time dynamics.

Let the pre-confinement dark-radiation energy density immediately before confinement be $\rho_{dr,D}(a_c^-)$. A general transition bookkeeping may be written as

$$\rho_{dr,D}(a_c^-) = \rho_{X,c}^{\text{init}} + V_{D,\text{eq}}^{\text{init}} + \rho_{m,D}^{\text{init}}, \quad (61)$$

with fractional decomposition

$$f_X + f_{\text{eq}} + f_m = 1, \quad \rho_{X,c}^{\text{init}} = f_X \rho_{dr,D}(a_c^-), \quad V_{D,\text{eq}}^{\text{init}} = f_{\text{eq}} \rho_{dr,D}(a_c^-). \quad (62)$$

Here f_X feeds the sourced arrested tail, f_{eq} feeds the exact-equilibrium remainder that returns to the equilibrium baseline, and f_m would describe any long-lived massive hidden relic component. The minimal version studied in this draft sets

$$f_m = 0, \quad (63)$$

and uses a single post-transition amplitude to encode the sourced branch,

$$\Sigma(a_c^+) = \varepsilon_\Sigma \Lambda_D^4, \quad \rho_X(a_c^+) = 3c_D \varepsilon_\Sigma \Lambda_D^4. \quad (64)$$

The exact-equilibrium remainder is then understood to return to $T_{\mu\nu}^{\text{eq}}$, not to survive as a separately gravitating FRW component.

If $\beta = 0$, the source-free branch is

$$\rho_X(a) = 3c_D \varepsilon_\Sigma \Lambda_D^4 \left(\frac{a}{a_c} \right)^{-1/c_D}. \quad (65)$$

For production Boltzmann runs it is usually safer to replace a sharp activation by the smooth profile

$$S(a) = \frac{1}{2} \left[1 + \tanh \left(\frac{\ln a - \ln a_c}{\Delta_c} \right) \right], \quad \Delta_c \ll 1. \quad (66)$$

Using $N \equiv \ln a$, a practical background implementation is

$$\frac{d\rho_c}{dN} = -3\rho_c - S(a) \frac{\beta}{c_D} \Psi(r) \rho_X, \quad (67)$$

$$\frac{d\Sigma}{dN} = -\frac{1 - \beta S(a) \Psi(r)}{c_D} \Sigma + \Sigma_c \frac{dS}{dN}, \quad \Sigma_c \equiv \varepsilon_\Sigma \Lambda_D^4, \quad (68)$$

$$\rho_X = 3c_D \Sigma, \quad r = \frac{\rho_c}{\rho_X}. \quad (69)$$

If the pre-confinement dark-radiation bath is to be switched off explicitly in the sourced subsystem, one may also use

$$\rho_{dr,D}(a) \rightarrow [1 - S(a)] \rho_{dr,D}(a). \quad (70)$$

Equations (67)-(70) are an EFT activation prescription for the post-confinement tail. They are not claimed to constitute a microphysical model of the phase transition.

5.2. Microscopic reconstruction and background-only benchmarks

The phenomenological late-time parameters map back to the microscopic picture through a short chain of relations. From Eq. (15),

$$\xi = \left(\frac{\Delta N_{\text{eff}}}{13.2} \right)^{1/4}. \quad (71)$$

Then Eq. (17) gives

$$\Lambda_D = \frac{\xi T_{\gamma,0}}{x_c} (1 + z_c). \quad (72)$$

The ultraviolet gauge coupling is

$$\alpha_D(M_{\text{Pl}}) = \frac{2\pi}{b_0 \ln(M_{\text{Pl}}/\Lambda_D)}, \quad b_0 = \frac{22}{3}. \quad (73)$$

Finally,

$$\varepsilon_\Sigma = \frac{\rho_X(a_c^+)}{3c_D \Lambda_D^4}. \quad (74)$$

In the source-free case,

$$\rho_X(a_c^+) = \rho_{X,0} (1 + z_c)^{1/c_D}. \quad (75)$$

In the signed-kernel case,

$$\rho_X(a_c^+) = \rho_{X,0} \exp \left[\int_{a_c}^1 \frac{1 - \beta \Psi[r(a)]}{c_D} d \ln a \right]. \quad (76)$$

This integral should be evaluated numerically on the sampled background.

The microscopic interpretation only survives if the preferred phenomenological region maps back to sensible values of $(\alpha_D(M_{\text{Pl}}), \xi, x_c, \varepsilon_\Sigma)$. Three of these four are set by the transition inputs $(\Delta N_{\text{eff}}, z_c, x_c)$, which were held fixed in the Phase 3 chains (Table 5) rather than sampled. The 3b input `yaml` passes no `Omega0_dr_dark` and no `z_conf`, so the runs use the `CLASS_SYMT` defaults: dark radiation switched off ($\Delta N_{\text{eff}} \approx 0$) and the late confinement redshift $z_c \approx 10^5$; x_c has no sampled counterpart. In this configuration the chain has *no* purchase on ξ , Λ_D or $\alpha_D(M_{\text{Pl}})$, because the relations (71)–(73) fold to limits of zero (or to ill-defined logarithms) at $\Delta N_{\text{eff}} = 0$. Consequently ξ , x_c , Λ_D and $\alpha_D(M_{\text{Pl}})$ carry no posterior from the present chains. At the *published illustrative benchmark* of Eq. (78) $(\Delta N_{\text{eff}}, z_c, x_c) = (10^{-4}, 200, 1)$ they take the fixed values $\xi = 0.052$, $\Lambda_D = 2.48 \text{ meV}$ and $\alpha_D(M_{\text{Pl}}) = 0.0124$, consistent with the dimensional-transmutation estimate of Eq. (12); these benchmark values are not the inputs the 3a/3b/3c chains actually used (which had $\Delta N_{\text{eff}} = 0$ and $z_c \approx 10^5$) but are quoted here to illustrate the reconstruction at a physically representative point in the un-sampled subspace. A genuine posterior on these three requires the growth-extension chains to sample $(\Delta N_{\text{eff}}, z_c, x_c)$ directly.

The one reconstruction amplitude that the 3b chain does inform is ε_Σ , which inherits the posterior on c_D (and, weakly, Ω_X) at fixed Λ_D through the source-free relation (75). Propagating the 3b samples gives

$$\varepsilon_\Sigma = 0.012_{-0.008}^{+0.023} \quad (68\%), \quad \varepsilon_\Sigma < 0.061 \quad (95\%), \quad (77)$$

a right-skewed distribution inherited from the broad, prior-dominated c_D marginal. The order- 10^{-2} amplitude is a sensible hidden-sector initial pressure deficit, so the reconstruction is not hollow; it is simply that, on the present stack, only ε_Σ is data-constrained while (α_D, ξ, x_c) remain fixed by un-sampled inputs.

Before any CLASS or MCMC analysis, the background system can already be made concrete. We therefore present two illustrative benchmark backgrounds obtained by backward-integrating Eqs. (44) and (45) from $a = 1$ to $a_c = (1 + z_c)^{-1}$, using

$$(\Omega_{b,0}, \Omega_{c,0}, \Omega_{r,0}, \Omega_{X,0}) = (0.049, 0.261, 9.2 \times 10^{-5}, 0.689908), \quad (78)$$

Quantity	3b result	Status
ξ	0.052	fixed (ΔN_{eff} not sampled)
x_c	1	fixed (no sampled parameter)
Λ_D	2.48 meV	fixed ($z_c, \Delta N_{\text{eff}}, x_c$ not sampled)
$\alpha_D(M_{\text{Pl}})$	0.0124	fixed (follows Λ_D)
ε_Σ	$0.012^{+0.023}_{-0.008}$ (< 0.061 , 95%)	3b-informed (via c_D)

Table 2: Microscopic reconstruction from the 3b posterior. Only ε_Σ is constrained by the chain; the remaining three are fixed by the transition inputs ($\Delta N_{\text{eff}}, z_c, x_c$) that Phase 3 held fixed, and are quoted at the benchmark of Eq. (78).

flatness at $a = 1$, $x_c = 1$, and the common transition inputs $z_c = 200$ and $\Delta N_{\text{eff}} = 10^{-4}$. These are *illustrative* backgrounds, not a fit to data.

Benchmark	z_c	ΔN_{eff}	c_D	β	w_X	$w_{\text{eff}}^{\text{min}}$	ξ	Λ_D [meV]	$\alpha_D(M_{\text{Pl}})$	ε_Σ
A	200	10^{-4}	8.0	1.5	-0.9583	-0.9824	0.0525	2.476	0.0124	0.0519
B	200	10^{-4}	5.5	3.0	-0.9394	-1.0094	0.0525	2.476	0.0124	0.0937

Table 3: Illustrative background benchmark points obtained by backward integration from $a = 1$. Benchmark A is the non-crossing branch. Benchmark B crosses below $w_{\text{eff}} = -1$ only mildly, between $z \approx 0.96$ and $z \approx 1.79$, while remaining intrinsically non-phantom. Its $\beta = 3.0$ lies inside the analytic crossing strip (102) but outside the narrow 3c sampling gate of Table 5; the point is chosen to display the crossing morphology, not to represent a sampled configuration.

Figure 1 shows the resulting background evolution. Benchmark A stays on the non-crossing branch. Benchmark B enters a mild effective phantom phase, but only at the level of $\sim 10^{-2}$ below -1 , exactly as anticipated from Eq. (52). The exchange term changes sign near $r = 1$, and the expansion rate differs from the corresponding Λ CDM benchmark by at most a few percent over the range shown.

These plots do not replace a Boltzmann analysis, but they are the right pre-CLASS benchmark: they make the background structure tangible, they show the sign logic explicitly, and they reveal that the allowed effective phantom phase is expected to be shallow rather than dramatic.

5.3. Linear perturbations

Why the stock fld module is not the theory. The theory contains a genuine memory variable, Σ . It is therefore not equivalent to specifying a back-

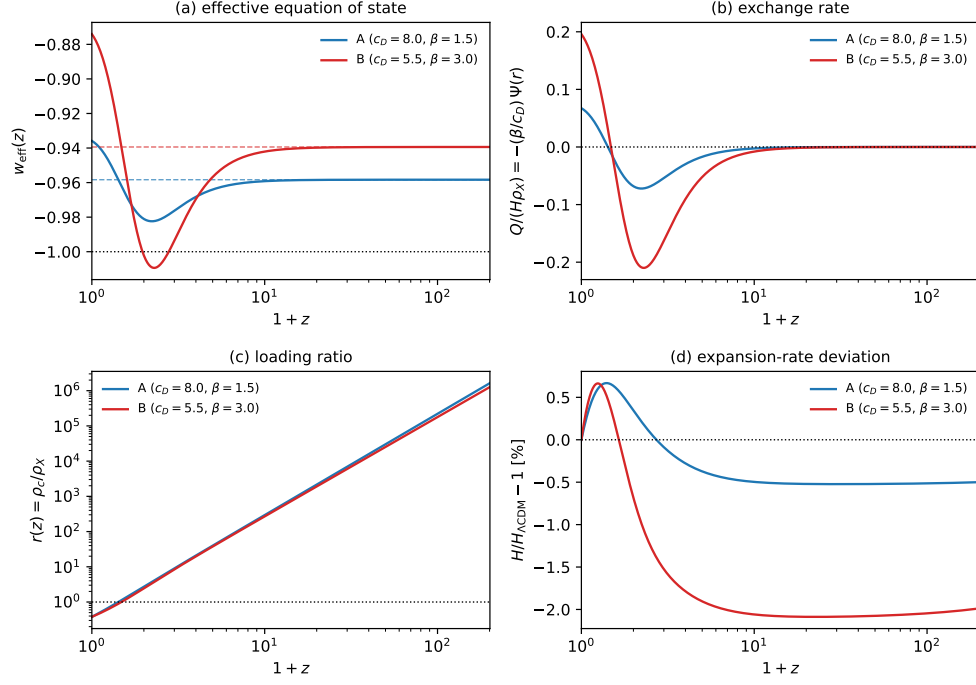


Figure 1: Background-only benchmark evolution for the two illustrative points in Table 3. The dashed horizontal lines in panel (a) show the intrinsic constant values w_X for the two benchmarks; the dotted line marks $w = -1$. Panel (b) shows the sign-changing exchange rate, panel (c) the loading ratio $r(z)$, and panel (d) the deviation of the Hubble rate from a flat Λ CDM model with the same present-day $(\Omega_{b,0}, \Omega_{c,0}, \Omega_{r,0}, \Omega_{X,0})$.

Symbol	Definition	Role
Σ	$\rho_X = 3c_D \Sigma$	background memory variable (Israel-Stewart)
s	$s \equiv \delta \Sigma$	independent perturbation DOF; evolved, not reconstructed
σ	$\sigma \equiv \delta \Sigma / \bar{\rho}_X$	normalised memory perturbation (code variable <code>sigma</code>)
δ_X	$\delta_X \equiv \delta \rho_X / \bar{\rho}_X$	density contrast of the afterglow fluid
δp_X	Eq. (84)	pressure perturbation (non-adiabatic: tied to σ , not δ_X)
ν_X	$\nu_X \equiv (\bar{\rho}_X + \bar{p}_X) \theta_X = \bar{\Sigma} \theta_X$	momentum (velocity-weighted) variable
χ	$\chi \equiv \delta \rho_X - 3c_D s$	non-adiabatic mismatch (need not vanish)

Table 4: Variables used in the linear perturbation sector. The defining feature of the closure is that $s = \delta \Sigma$ is an *independent* degree of freedom: the non-adiabatic mismatch χ measures the departure from the algebraic sub-manifold $\delta \rho_X = 3c_D s$ and evolves dynamically rather than being set to zero.

ground equation of state $w(a)$ and passing that function to the stock CLASS generalized-fluid module. Doing so would collapse a causal relaxation system into an instantaneous barotropic closure. That is not the model defined above. A correct Boltzmann treatment must evolve the perturbation of the constitutive variable explicitly.

Background-exact identity versus independent memory mode. The most important conceptual clarification is simple. The relation

$$\bar{\rho}_X = 3c_D \bar{\Sigma} \quad (79)$$

is an exact *homogeneous FRW identity* of the post-transition arrested branch. It is not a local algebraic constitutive law that forces perturbations to satisfy $\delta\rho_X = 3c_D\delta\Sigma$ at every spacetime point.

Accordingly, in linear theory the perturbation of the memory variable is independent. Define

$$s \equiv \delta\Sigma, \quad \chi \equiv \delta\rho_X - 3c_D s. \quad (80)$$

The quantity χ measures the non-adiabatic mismatch between the density perturbation and the perturbation of the memory variable. In this framework χ need not vanish. The correct perturbative question is therefore not whether $\delta\rho_X$ can be reconstructed algebraically from s , but how χ evolves under the constitutive closure.

Baseline frame and dynamical variables. The signed-kernel background law does not by itself fix the perturbative transfer frame. For the first explicit implementation we adopt the standard baseline choice

$$Q^\nu = Q u_c^\nu, \quad (81)$$

which closes the scalar sector directly in standard synchronous gauge. This is a baseline implementation choice, not an additional response shell. Once the baseline system is stable, a re-derivation in the Landau frame of the combined dark sector is a worthwhile robustness check.

We work in synchronous gauge,

$$ds^2 = a^2(\eta) [-d\eta^2 + (\delta_{ij} + h_{ij})dx^i dx^j], \quad (82)$$

and choose the usual cold-dark-matter comoving gauge so that $\theta_c = 0$. Because the baseline transfer vector is aligned with u_c^μ , that condition is preserved.

For the sourced tail define the normalized variables

$$\delta_X \equiv \frac{\delta\rho_X}{\bar{\rho}_X}, \quad \sigma \equiv \frac{\delta\Sigma}{\bar{\rho}_X}, \quad (83)$$

and the velocity divergence θ_X ; the momentum variable is $\nu_X \equiv \bar{\Sigma} \theta_X$.

Stabilised fluid closure. The perturbation closure implemented in the Phase 3 code and used for all numerical results in this paper is a CLASS-fluid-style stabilized system. The dark-energy fluid uses a rest-frame sound speed $c_{s,X}^2$ (named `cs2_X` in the `afterglow_pert.c` module), which enters the standard fluid continuity and Euler equations. The pressure perturbation is set by the stabilized closure

$$\delta p_X = c_{s,X}^2 \bar{\rho}_X \delta_X + w_X \bar{\rho}_X (\sigma - \delta_X). \quad (84)$$

This is *not* the naïve barotropic closure $\delta p_X = w_X \delta\rho_X$. The second term ensures that the non-adiabatic content of the memory variable enters the pressure perturbation. For comparison, the constitutive relation $p_X = -\rho_X + \Sigma$ would give the bare closure

$$\delta p_X = -\delta\rho_X + s, \quad (85)$$

to which Eq. (84) reduces when the $(c_{s,X}^2 - w_X)$ stabilization is removed. We retain the stabilized form because it is the closure that was integrated in the Phase 3 code; the bare form (85) is the natural starting point for a future theory-first closure revision (see remark below).

The evolution system is

$$\begin{aligned} \delta'_X = & -(1 + w_X) \left(\theta_X + \frac{1}{2} h' \right) - 3\mathcal{H} (c_{s,X}^2 - w_X) \delta_X \\ & - 9(1 + w_X) (c_{s,X}^2 - w_X) \mathcal{H}^2 \theta_X / k^2 \\ & - 3\mathcal{H} w_X (\sigma - \delta_X) + \mathcal{H} (\beta / c_D) \delta\Psi, \end{aligned} \quad (86)$$

$$\theta'_X = -\mathcal{H} (1 - 3c_{s,X}^2) \theta_X + \frac{c_{s,X}^2 k^2}{1 + w_X} \sigma, \quad (87)$$

$$\begin{aligned} \sigma' = & \frac{\mathcal{H}}{c_D} \beta r \Psi'(r) (\delta_c - \delta_X) \\ & - \frac{\theta_X + \frac{1}{2} h'}{3 c_D} [1 - \beta \Psi(r)]_{\text{eff}}. \end{aligned} \quad (88)$$

Here $\delta\Psi$ is the linear response of the kernel to the density contrasts,

$$\delta\Psi \equiv r \Psi'(r) (\delta_c - \delta_X), \quad (89)$$

the same combination that drives the memory equation (88).

Pressure-transfer bookkeeping. One internal feature of the stabilized system should be stated explicitly. The pressure perturbation entering the continuity equation (86) is the full stabilized closure (84), while the pressure gradient entering the Euler equation (87) is built from the memory variable alone, $\delta p_X^{(\nabla)} = c_{s,X}^2 \bar{\rho}_X \sigma$. The two agree on adiabatic configurations ($\delta_X = \sigma$) and differ otherwise by a non-adiabatic term $(c_{s,X}^2 - w_X)(\delta_X - \sigma) k^2 / (1 + w_X)$ in the Euler equation. This asymmetric bookkeeping is inherited from the CLASS fluid template on which the module is built, and it is part of the closure freedom already disclosed in this section; the two-closure reconciliation planned for Phase 3.5 removes it together with the c_b^2 question. The $[1 - \beta\Psi(r)]_{\text{eff}}$ notation in (88) denotes the smooth B_1 regulator $f_{\text{reg}}(x, \delta) = \frac{1}{2}(x + \sqrt{x^2 + \delta^2})$ with $\delta = 10^{-4}$, applied to $1 - \beta\Psi$ to absorb the sign flip at $\beta\Psi(r) > 1$ inside the crossing strip (102).

The $(c_{s,X}^2 - w_X)$ terms in (86) provide super-horizon damping that prevents gradient instabilities in the CLASS integrator. The $c_{s,X}^2 / (1 + w_X)$ coefficient in (87) sets the pressure-gradient response. In the Phase 3 chains $c_{s,X}^2 = 0.5$ was held fixed; the Phase 3.5 analysis will sample it.

Perturbed interaction. For the signed-kernel exchange law (42), the perturbation follows by varying the same local variables that define it. In the cold-dark-matter rest frame,

$$a \delta Q = -\beta\Psi(r) \left(3\mathcal{H}s + \bar{\Sigma} \frac{h'}{2} \right) - 3\beta\mathcal{H}\bar{\Sigma} \Psi_{,r} \delta r, \quad (90)$$

with

$$\delta r = \frac{1}{\bar{\rho}_X} (\delta\rho_c - r \delta\rho_X), \quad \Psi_{,r} = \frac{4(-r^2 + 4r - 1)}{(1+r)^4}. \quad (91)$$

The perturbed exchange enters (86) through the $\delta\Psi$ term and (88) through the $\beta r \Psi'(r)(\delta_c - \delta_X)$ source.

Bare closure, short-wave limit, and the c_b^2 alternative. For theoretical transparency it is useful to record the bare (un-stabilized) closure that follows directly from the constitutive relation. In terms of the un-normalized variables $(\delta\rho_X, \nu_X, s)$, linearising $\nabla_\mu T_c^{\mu\nu} = Q^\nu$ and $\nabla_\mu \tau_X^{\mu\nu} = -Q^\nu$ in the baseline frame gives

$$\delta\rho'_c + 3\mathcal{H}\delta\rho_c + \bar{\rho}_c \frac{h'}{2} = a \delta Q, \quad (92)$$

$$\delta\rho'_X + 3\mathcal{H}s + \nu_X + \bar{\Sigma}\frac{h'}{2} = -a\delta Q, \quad (93)$$

$$\nu'_X + \left[4 - 3\beta\Psi(r)\right]\mathcal{H}\nu_X - k^2(-\delta\rho_X + s) = 0. \quad (94)$$

These are exact consequences of the conservation equations and the exchange law; they do not depend on the closure choice. (Equation (94) uses $a\bar{Q} = -3\beta\mathcal{H}\Psi(r)\bar{\Sigma}$.)

An alternative memory-mode closure, not used in the Phase 3 code but recorded here as a theoretical target for future work, is

$$s' = -\frac{\mathcal{H}}{c_D}s - \frac{a\delta Q}{3c_D} - (1 + c_b^2)\left(\nu_X + \bar{\Sigma}\frac{h'}{2}\right), \quad 0 < c_b^2 \leq 1. \quad (95)$$

Here c_b^2 is a bulk-mode dispersion coefficient that controls the short-wavelength propagation speed of the memory mode. In the short-wavelength, no-expansion, no-interaction limit ($\mathcal{H} = \bar{Q} = h' = 0$), the bare system gives

$$\delta\rho'_X + \nu_X = 0, \quad \nu'_X - k^2(-\delta\rho_X + s) = 0, \quad s' = -(1 + c_b^2)\nu_X. \quad (96)$$

For plane waves $\propto e^{-i\omega\eta}$ this yields

$$\omega^2 = c_b^2 k^2. \quad (97)$$

Thus the bare closure has a clean hyperbolic short-wave limit with propagation speed c_b^2 . The stabilized code closure shares this property with $c_{s,X}^2$ playing the analogous role; the two closures differ primarily in their super-horizon damping structure. Reconciling them — either by adding c_b^2 -style memory dispersion to the code, or by deriving the stabilization terms from the constitutive relation directly — is a planned revision for the Phase 3.5 code. The deferral is verified not to affect the present constraints: the data are insensitive to the rest-frame sound speed across $c_{s,X}^2 = 0.25$ and $c_{s,X}^2 = 1.20$, a range far wider than the closure discrepancy at issue, leaving the reported bounds unchanged within sampling error. Throughout, the rest-frame sound speed is held strictly positive, so the perturbation sector respects the $c_s^2 \geq 0$ stability condition on which the single-field no-go results rest [4, 5]; the apparent crossing of the effective equation of state is a two-component transfer effect, exactly the multi-component route those results leave open. The Phase 3 numerical results reported in this paper use exclusively the stabilized closure (84) with the evolution system (86)–(88).

Baseline-frame crossing strip and practical guards. The explicit synchronous-gauge system immediately yields the most important practical bound; guards of this kind are the afterglow analogue of the large-scale instability conditions familiar from coupled dark-energy fluids [29]. Equation (94) shows that the local momentum-friction coefficient of the sourced tail is

$$\Gamma_\nu = \left[4 - 3\beta\Psi(r)\right]\mathcal{H}. \quad (98)$$

Since $\Psi(r) < 0$ for $r < 1$, the dark-energy-dominated regime is automatically safe in the explicit baseline frame. Anti-damping is possible only in the matter-loaded regime $r > 1$, where $\Psi(r) > 0$.

Transient effective phantom crossing requires $\beta\Psi(r) > 1$. The signed kernel reaches its positive maximum at $r = 2 + \sqrt{3}$, with

$$\Psi_{\max} = \frac{2\sqrt{3}}{9}. \quad (99)$$

Therefore crossing is possible only if

$$\beta > \frac{1}{\Psi_{\max}} = \frac{3\sqrt{3}}{2} \approx 2.60. \quad (100)$$

To avoid local anti-damping in the same regime one needs $\beta\Psi(r) < 4/3$. Using Eq. (99), a sufficient global guard is

$$\beta < \frac{4}{3\Psi_{\max}} = 2\sqrt{3} \approx 3.46. \quad (101)$$

Hence the explicit baseline implementation supports a baseline-frame crossing strip,

$$\frac{3\sqrt{3}}{2} < \beta < 2\sqrt{3}. \quad (102)$$

The remaining stability guards are straightforward:

1. $c_D > 1/2$ if the intrinsic sourced branch is to accelerate;
2. $c_D > (1 + 4\beta)/3$ if the late-time decaying branch $r \rightarrow 0$ is to be stable;
3. $\bar{\Sigma} > 0$ and $\bar{\rho}_X = 3c_D\bar{\Sigma} > 0$ at all times;
4. $0 < c_{s,X}^2 \leq 1$ so the short-wave system is hyperbolic and subluminal;

5. $\beta \geq 0$.

Within these guards, the model can support an effective crossing branch without ever making the intrinsic dark-energy sector phantom. The benchmark of Section 5 shows that the resulting excursion is typically mild, which is the physically relevant expectation to carry into the first CLASS runs.

5.4. Numerical implementation and analysis pipeline

The observational program was staged. The exact $\beta = 0$ branch was validated first, at the background level and then with the full perturbation sector, before the signed kernel was confronted with production chains; the interacting analysis was then run as two separate branches, below and inside the explicit baseline crossing strip (102). Table 5 defines the three resulting Phase 3 chains. The staged bring-up sequence, the pre-run validation checklist together with its verification status, and a recommended prior box for the growth-extension chains are collected in Appendix F.

Branch	β treatment	Sampled parameters	Samples	$R - 1$
3a	$\beta = 0$ (isolated tail)	standard Λ CDM set + $\log_{10} c_D$	5.1×10^5	6.7×10^{-3}
3b	free, non-crossing, $\beta \in (0, 2.4]$	standard Λ CDM set + $\log_{10} c_D, \beta$	7.6×10^5	5.4×10^{-3}
3c	free, crossing (narrow gate), $\beta \in [2.65, 2.75]$	standard Λ CDM set + $\log_{10} c_D, \beta$	1.3×10^6	5.6×10^{-3}

Table 5: Definition of the three Phase 3 chains. All branches share the likelihood stack Planck NPIPE CamSpec TTTEEE + low ℓ .TT + low ℓ .EE_{sroll2} + Planck lensing + DESI DR2 BAO + Pantheon+, sampled with Cobaya [33] using fast/slow blocking and four chains per branch, and share the fixed inputs $c_{s,X}^2 = 1/2$, $\Delta N_{\text{eff}} \approx 0$ (pre-confinement dark radiation switched off), $z_c \approx 10^5$ (the CLASS_SYMT default), with no sampled counterpart for x_c ; the source amplitude is fixed by the flatness shooting condition described below. The crossing-branch prior sits inside the baseline strip (102), and every accepted sample satisfies the late-branch guard (55). Sample counts are post-burn-in accepted-state counts after a 30% burn-in cut; $R - 1$ is the Gelman–Rubin value at chain stop.

Pipeline. The chains were produced with CLASS_SYMT [31] (https://github.com/lawdroid/CLASS_SYMT), a modification of the public CLASS

solver [32] implementing the background and perturbation equations of Sections 4–5, driven by the `Cobaya` MCMC sampler [33] with fast/slow blocking on four chains per branch. Convergence was assessed with the Gelman–Rubin statistic; final values and post-burn-in accepted-state counts, after a 30% burn-in cut, are listed in Table 5. The rest-frame sound speed of the code closure was held at $c_{s,X}^2 = 1/2$ throughout Phase 3 (the parameter is named `cs2_X` in the input `yaml`); the growth-extension analysis will sample it. The perturbation equations integrated by the code are Eqs. (86)–(88) of Section 5.3; the alternative c_b^2 closure of Eq. (95) is recorded as a theoretical target and was not used in the Phase 3 runs. Chain files, run logs, input `yaml`, and a per-branch `CHAINS_TECHNICAL.md` companion document are archived together with the figures in this section’s source bundle.

Background sector. `CLASS_SYMT` adds a dedicated sourced-tail species with background quantities $\{\Sigma, \rho_X, r, \Psi, Q\}$, plus the optional pre-confinement dark-radiation component $\rho_{dr,D}$ and the switch $S(a)$. The implementation evolves Σ and derives $\rho_X = 3c_D \Sigma$ algebraically, which preserves the defining *background* identity exactly by construction. Two amplitude strategies are implemented. Backward integration starts at $a = 1$ with

$$\rho_{X,0} = 3M_{\text{Pl}}^2 H_0^2 - \rho_{b,0} - \rho_{c,0} - \rho_{r,0} - \rho_{dr,D,0}, \quad (103)$$

sets $\Sigma_0 = \rho_{X,0}/(3c_D)$, and integrates backward to reconstruct $\rho_X(a_c^+)$ and ε_Σ ; it was used to verify that the background equations are coded correctly and to produce the benchmark backgrounds of Table 3. Production runs integrate forward, using Σ_c (equivalently ε_Σ) as a shooting parameter adjusted until flatness at $a = 1$ is satisfied, with the source-free analytic estimate of Eq. (75) as the initial guess.

Perturbation sector. The code adds the variables $\{\delta_X, \theta_X, \sigma\}$ and evolves the stabilized system (86)–(88) together with the exchange perturbation (90). The defining implementation rule is that $\sigma = \delta\Sigma/\bar{\rho}_X$ is never reconstructed algebraically from δ_X : the independent perturbation of the constitutive variable is the whole point of the model. The non-adiabatic mismatch $\chi = \delta\rho_X - 3c_D s$ of Eq. (80) is monitored as an evolved diagnostic; the model predicts that χ evolves dynamically, and $\chi \neq 0$ is not a numerical error.

Configuration and initial conditions. The Phase 3 chains run in the phenomenological basis: the standard Λ CDM parameters together with $\log_{10} c_D$

(and, on the interacting branches, β), with the source amplitude fixed by the flatness shooting condition. The transition inputs were held at the `CLASS_SYMT` defaults (dark radiation switched off, $\Delta N_{\text{eff}} \approx 0$, and activation at $z_c \approx 10^5$), so the microscopic parameters $(\xi, \Lambda_D, \alpha_D, x_c)$ carry no posterior on the present stack (Section 5.2). Sampling $(z_c, \Delta N_{\text{eff}})$ directly is deferred to the growth-extension chains, for which either the microscopic basis $\{\alpha_D(M_{\text{Pl}}), \xi, x_c, \varepsilon_\Sigma, c_D, c_{s,X}^2, \beta\}$ or the numerically better-conditioned phenomenological basis $\{z_c, \Delta N_{\text{eff}}, c_D, c_{s,X}^2, \beta\}$ may be used, with the microscopic reconstruction performed afterward as in Section 5.2. When the transition occurs early with an active dark-radiation bath, adiabatic initial conditions for the pre-confinement bath are sufficient [34] and the switch generates the tail dynamically. For sharp-switch validation the matched post-transition choice

$$s(a_c^+) = \frac{1}{3c_D} \delta\rho_X(a_c^+), \quad \nu_X(a_c^+) = 0 \quad (104)$$

was used; it is a controlled initialization for code validation rather than a unique physical prescription.

Branch structure.. The signed-kernel interaction is not sampled on the full positive real line. The two interacting branches sample $\beta \in (0, 2.4]$ (non-crossing, 3b) and $\beta \in [2.65, 2.75]$ (crossing, 3c), respectively below and inside the explicit baseline strip (102). The 3c prior is the narrow entry gate of the crossing strip $\beta \in [2.65, 3.40]$: its upper edge is set just below the region where the B_1 regulator materially deforms the dynamics, so that the sampled posterior is shaped by the data rather than by regulator damping. Every accepted sample additionally satisfies the late-branch guard (55), enforced inside the integrator and monitored as the derived chain parameter `late_branch_guard`.

Validation.. The background and perturbation sectors are exercised by a unit suite that mirrors the production C code line-for-line and asserts the analytic limits of Sections 4–5: on commit `b9d78d9` the suite runs 65 assertions (38 background, 27 perturbation) with zero failures, and the linear-order background Bianchi residual is 8.5×10^{-17} . The eleven pre-run checks, the mapping from each check to the test that establishes it, and the quantitative results are collected in Appendix F (Table F.6); the two integration-level checks (full-likelihood runs on the $\beta = 0$ branch before the interacting branches, and convergence of the spectra with respect to the switch width

Δ_c and the k -sampling of the new species) were satisfied by the Phase 3 production runs themselves.

5.5. Constraints on the data-resolved combination $g \equiv \beta/c_D$

Because β appears in the equations of motion only through the ratio $g \equiv \beta/c_D$ (Eqs. (42)–(45)), while c_D appears alone in w_X (geometry) and in the $1/c_D$ prefactor on the memory-mode evolution (88) (perturbations), the non-crossing chains constrain c_D directly but leave the β marginal flat across its prior. The Savage–Dickey density ratio makes the equivalence explicit: a flat marginal in the β basis and a bounded marginal in the g basis are the same statement. Operationally we recommend post-processing each chain in the g basis, $g_i = \beta_i/c_{D,i}$, and reporting g_{95} as the data-driven coupling constraint.

From the Phase 3b non-crossing chain (Planck NPIPE CamSpec TT+TEEE + low ℓ .TT + low ℓ .EE_{scroll2} + Planck lensing + DESI DR2 BAO + Pantheon+; four chains; 7.6×10^5 post-burn-in accepted states) the resulting posterior is shown in Fig. 2. The summary statistics are

$$g_{50} = 0.038, \quad g_{84} = 0.115, \quad g_{95} = 0.195, \quad (105)$$

so the geometry+CMB stack bounds the physical coupling at $g < 0.20$ at 95% credibility while the β marginal remains featureless across $\beta \in (0, 2.4]$. A direct test of g in its defining channel — the growth modulation entering $\sigma_8(z)$ and $f\sigma_8(z)$ — requires extending the stack with redshift-space distortions (e.g. DESI Y3 full-shape) and cosmic shear (e.g. DES Y3 3×2 pt), which is deferred to a separate analysis.

Identifiability of c_D . The structural claim that c_D is solo-pinned by its β -free terms can be stress-tested directly against the chains. We compare the marginal posterior for c_D from the $\beta = 0$ branch (Phase 3a) against that from the β -free branch (Phase 3b); see Fig. 3. The posterior means shift by

$$\Delta\langle c_D \rangle = \langle c_D \rangle_{3b} - \langle c_D \rangle_{3a} \approx +1.14, \quad (106)$$

on a posterior scale of order ~ 30 , while a two-sample Kolmogorov–Smirnov test on draws thinned to the c_D effective sample size of each chain ($N_{\text{eff}} \approx 2.7 \times 10^3$ and 2.3×10^3 for 3a and 3b) gives $D_{\text{KS}} = 0.029$ with $p = 0.22$: at the independence-corrected resolution of the chains, the two marginals are statistically indistinguishable. The shift is small in absolute terms ($\lesssim 5\%$

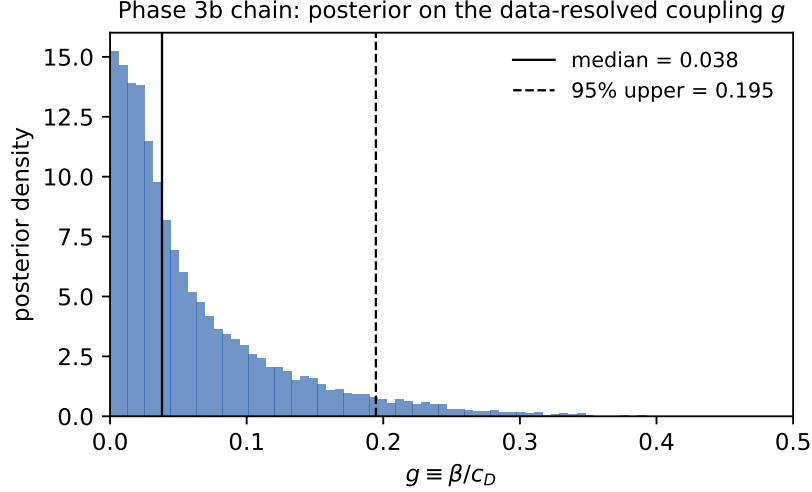


Figure 2: Posterior on the data-resolved coupling $g \equiv \beta/c_D$ from the Phase 3b non-crossing chain. Solid line marks the weighted median $g_{50} = 0.038$; dashed line marks the 95 % upper bound $g_{95} = 0.195$.

of the median, well within the 68 % width of the 3a marginal) and is not resolved as significant by the distribution-level test. The interpretation: c_D is dominantly pinned by its solo terms, with any residual $\mathcal{O}(\beta\Psi)$ coupling through the perturbation closure entering below the resolution of the present stack. The framing “ c_D solo-pinned, g flat” therefore holds at the level the chain resolves on the present stack, with a quantitative residual recorded above for completeness.

The value of c_D itself deserves one structural remark. The regime $c_D \gtrsim 5$ was specified in advance by the attractor and crossing-guard conditions of Sections 4–5, and the data land inside it; the value is fitted. No flat-space transport mechanism produces a relaxation time of order c_D/H : lattice Kubo estimates of the bulk channel in confining gauge theories give microscopic relaxation scales [35], shorter than $1/H$ by tens of orders of magnitude. Read this way, the Deborah number is a measurement of an intrinsically gravitational relaxation channel rather than a tuned fluid parameter; the systematic case is developed in a companion paper.

Parameter correlations. Figure 4 shows the joint posterior of the data-resolved coupling g together with c_D and the two structure-growth sum-

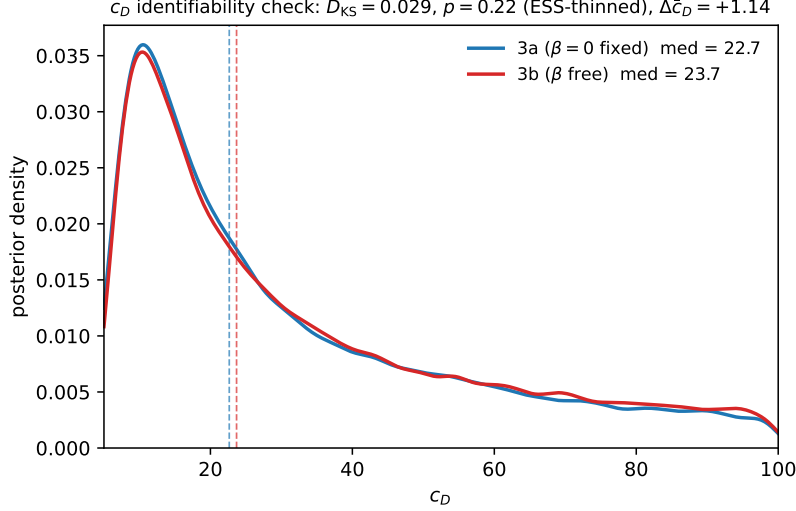


Figure 3: Devil’s-advocate check on the c_D identifiability claim. Marginal posterior on c_D from Phase 3a ($\beta = 0$ fixed) against Phase 3b (β free in $(0, 2.4]$). The two distributions overlay closely (mean shift $+1.14$ in c_D , $\lesssim 5\%$ of the median), confirming that c_D is dominantly constrained through its β -free terms rather than through indirect coupling.

maries Ω_m and σ_8 from the Phase 3b chain. The correlation structure makes the Savage–Dickey reasoning visual. The coupling is strongly anticorrelated with the late-time amplitude, $\text{corr}(g, \sigma_8) = -0.80$, and with the redshift-space growth rate, $\text{corr}(g, f\sigma_8) \simeq -0.75$ across $z \in [0.1, 0.5]$, while its anticorrelation with c_D (-0.54) is the algebraic shadow of $g = \beta/c_D$. The geometry+CMB stack used here pins Ω_m and σ_8 tightly ($\Omega_m = 0.304 \pm 0.005$, $\sigma_8 = 0.814 \pm 0.011$) and resolves the g – σ_8 degeneracy only through the CMB-anchored amplitude; a direct low-redshift measurement of σ_8 or $f\sigma_8$ would cut into the degeneracy along its short axis. This is precisely the channel in which g is defined, and it motivates the forecast below.

Growth-channel forecast for Phase 3.5. The present bound $g < 0.20$ is set indirectly — by the β prior cap combined with the c_D lower bound — because the geometry+CMB stack does not enter g ’s defining growth channel. A direct constraint requires redshift-space distortions and cosmic shear, the channel in which structure growth is the classic discriminator of energy-exchange couplings [36]. We estimate the gain with a single-parameter Fisher argument anchored on the chain’s own growth response; this linear extrapolation

Phase 3b non-crossing posterior: $(g, c_D, \Omega_m, \sigma_8)$

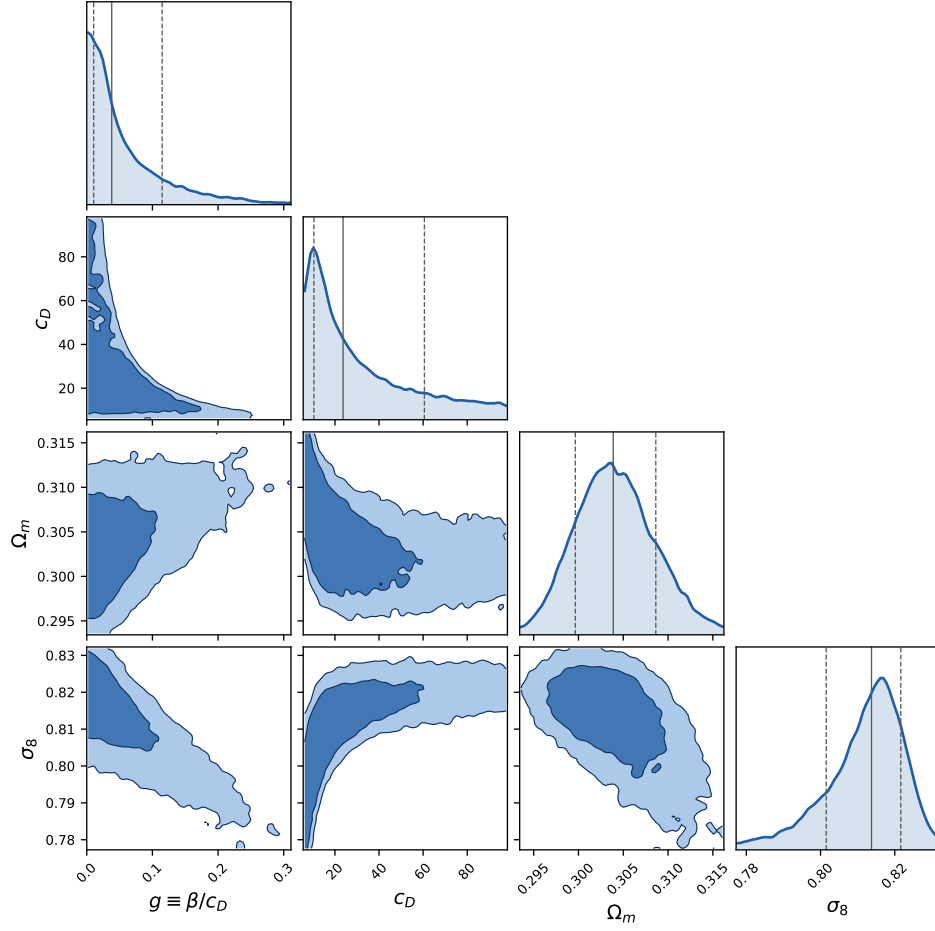


Figure 4: Triangle plot of the Phase 3b non-crossing posterior in the $(g, c_D, \Omega_m, \sigma_8)$ subspace, with $g \equiv \beta/c_D$ computed sample-by-sample. Diagonal panels are weighted 1D marginals (solid line, median; dashed lines, central 68 % interval); off-diagonal panels show the 68 % and 95 % credible regions. The strong g – σ_8 anticorrelation (-0.80) shows that growth is the channel that resolves g , visually confirming the Savage–Dickey argument of Sec. 5.5.

is a rough guide rather than a forecast, since the g posterior is right-skewed and bounded at zero. A profile-likelihood or mock-data injection analysis should replace it in the Phase 3.5 paper. From the Phase 3b samples the linear sensitivity of the low-redshift growth amplitude to the coupling is $\partial \ln f\sigma_8 / \partial g \simeq -0.15$ (equivalently $\partial \sigma_8 / \partial g|_{\Omega_m} \simeq -0.10$), so a measurement of the late-time growth amplitude to a fractional precision ϵ yields $\sigma_{\text{growth}}(g) \simeq \epsilon/0.15$. DESI Y3 full-shape $f\sigma_8(z)$ over $z \lesssim 1.5$ combined with the DES Y3 3×2 pt amplitude is forecast to reach $\epsilon \simeq 1\%$ on the growth amplitude [37, 38], giving $\sigma_{\text{growth}}(g) \simeq 0.07$. Added in quadrature to the present $\sigma_{\text{now}}(g) \simeq 0.06$, this sharpens the constraint to $\sigma(g) \simeq 0.046$ and roughly halves the 95% bound to $g \lesssim 0.10$ — now measured *through* the growth channel rather than inferred around it. A nonzero detection of g at the current posterior median ($g_{50} = 0.038$) would instead require $\sigma(g) \lesssim 0.02$, i.e. a Stage-IV growth-amplitude precision of $\epsilon \sim 0.3\%$ (Euclid, LSST, or the full DESI survey). This is the quantitative target for the Phase 3.5 chain, which will additionally sample $c_{s,X}^2$ rather than hold it fixed. The test is pre-registered rather than aspirational: the configuration (redshift-space distortions plus 3×2 pt likelihoods, freed $c_{s,X}^2$, sampled transition inputs) and its priors are fixed in advance in Appendix F, the growth data it requires are the DESI 2025–2026 full-shape and lensing releases, and DESI DR3 (expected 2027) is the dated horizon for the $w_X = -1 + 1/(3c_D)$ prediction.

Bayesian model comparison versus Λ CDM. Λ CDM is the nested limit $c_D \rightarrow \infty$ of every afterglow branch (equivalently $u \equiv 1 + w_X = 1/(3c_D) \rightarrow 0$), which makes the Bayes factor against Λ CDM a Savage–Dickey density ratio at the nested point, computable directly from the Phase 3 chains. Evaluating the ratio of the marginal posterior to the prior in u on each chain’s w_X column gives

$$\Delta \log Z (\text{afterglow} - \Lambda\text{CDM}) = \{+0.53 \pm 0.08, +0.35 \pm 0.06, +0.02 \pm 0.03\} \\ \text{for 3a, 3b, 3c,} \quad (107)$$

where the quoted uncertainty is dominated by a systematic of the method: the sampled prior caps u from below at $u \geq 1/(3c_D^{\text{max}}) > 0$, so the nested value $u = 0$ is reached only by extrapolating the (finite, well-behaved) posterior-to-prior ratio to the boundary. As an independent cross-check, a k -nearest-neighbour marginal-likelihood estimator [39] applied directly to the full-dimensional posterior samples — a method that shares no machinery

with the density-ratio construction — reproduces the *inter-branch* evidence differences to 0.001–0.015 in $\log Z$; the absolute anchor to the Λ CDM point is what carries the systematic above. All three values satisfy $|\Delta \log Z| < 1$, i.e. inconclusive on the Jeffreys scale, robustly across the boundary-extrapolation range. The small positive lean (favouring finite c_D over the $c_D \rightarrow \infty$ limit) is consistent with DESI DR2’s mild dynamical-dark-energy preference but does not rise to the level of significant Bayesian discrimination. A definitive, extrapolation-free PolyChord nested-sampling computation of both models under the same likelihood stack is staged for the same compute pipeline that will run Phase 3.5.

5.6. The coincidence question: a derived negative

The kernel’s zero at $r = 1$ invites an obvious question: does the interaction explain why matter and dark energy have comparable densities today? The answer, derived rather than assumed, is no. From the background equations (44)–(45), the density ratio $r = \rho_c/\rho_X$ obeys the autonomous flow

$$\frac{dr}{dN} = r \left[3w_X + g F(r) \right], \quad F(r) \equiv -\Psi(r) \frac{1+r}{r} = -\frac{4(r-1)}{(1+r)^2}, \quad (108)$$

with $N = \ln a$; the equation is an exact consequence of (44)–(45). Equality is a watershed of this flow rather than a resting point: $F(1) = 0$, so the trajectory crosses $r = 1$ at the uncoupled rate $\dot{r}/r = 3w_X H < 0$. The kernel vanishes where a coincidence mechanism would need it to act. A fixed point of (108) requires $g F(r_*) = -3w_X = 3 - 1/c_D$ with $F(r_*) > 0$, hence $r_* < 1$, and since $F < 4$ on $(0, 1)$ no fixed point of any kind exists below

$$g_* = \frac{3 - 1/c_D}{4} \in [0.70, 0.75] \quad \text{across the sampled prior } c_D \in [5, 100]. \quad (109)$$

The data-allowed region sits far below this threshold: g_* is 3.6 times the 95% bound $g_{95} = 0.195$ and roughly twenty times the posterior median $g_{50} = 0.038$. Even above g_* the stall would occur on the dark-energy side of equality ($r_* < 1$), because for $r > 1$ both terms of the bracket are negative; the interaction can never hold the universe at or before matter–dark-energy equality. The loading–unloading kernel is therefore centered on equality without solving coincidence, and the Phase 3 posterior confirms that the data do not ask it to.

6. Conclusion

The logic of the paper has two halves that now lock together.

1. In horizon thermodynamics, a Lorentz-invariant equilibrium vacuum contributes no null heat flow and therefore drops out of the macroscopic source law. Gravity responds to $\tau_{\mu\nu}$, not to the equilibrium baseline.
2. Because the universe is accelerating, there must exist a cosmic medium that remains persistently out of equilibrium and is still near vacuum today.
3. A hidden pure Yang-Mills sector provides the natural trigger: dimensional transmutation generates the meV scale, confinement supplies the thermodynamic shock, and the gapped post-transition phase supplies the slow relaxation.
4. The resulting afterglow is described by one memory variable Σ , one interaction amplitude β , one fixed sign-changing kernel $\Psi(r)$, and one independent scalar perturbation variable $\delta\Sigma$.

This reformulation is more economical than a generic interacting-fluid shell. The sourced tail itself is never phantom. What can cross -1 is only the *effective* uncoupled fit, and even then only mildly inside the baseline crossing strip. After dark-energy domination, the sign of Ψ reverses, Q becomes positive, and the effective equation of state becomes more quintessence-like than the intrinsic branch. Because the kernel dies away automatically for $r \rightarrow 0$ and $r \rightarrow \infty$, the far future is always the isolated decaying tail.

The framework is fully exposed to failure, and the first tier of tests has now been run. In the configuration tested, the perturbation sector is stable: the stabilized closure integrates, including on the crossing branch, and all three production chains converge (Table 5). The isolated $\beta = 0$ branch survives the full Planck+DESI DR2+Pantheon+ stack, the non-crossing interacting branch is bounded at $g < 0.20$ at 95% credibility without tension, and the one microscopic amplitude the chains inform, $\varepsilon_\Sigma \approx 0.01$, is a plausible hidden-sector initial pressure deficit. The remaining exposure is equally concrete. If the pre-registered growth test drives g to zero at the forecast precision, the interacting branch loses its empirical motivation. If the growth-extension chains, sampling $(\Delta N_{\text{eff}}, z_c, x_c)$ and $c_{s,X}^2$ directly, push the preferred region onto implausible microscopic values, the hidden-sector interpretation

fails even where the late-time effective fit succeeds. And within the thermodynamic reading of gravity, the proposal already does what it was built to do: it supplies the mandatory non-equilibrium source with a compact late-time effective theory that current data neither prefer nor exclude.

The $\beta = 0$ branch has cleared the first stack. The growth channel and the transition parameters are next, on a dated clock: the DESI 2025–2026 growth releases, and DR3 in 2027.

CRedit authorship contribution statement

Thomas G. Martin: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review and editing. **In-Gyu Koh:** Formal analysis, Investigation, Software, Validation, Visualization, Writing – original draft, Writing – review and editing.

Data availability

The afterglow modification of CLASS used in this work, CLASS_SYMT [31] (published on Zenodo, <https://doi.org/10.5281/zenodo.21301804>), together with the MCMC chains, run logs, input YAML files, and the per-branch technical companion documents <https://zenodo.org/records/21303930>), is openly available through releases hosted at https://github.com/lawdroid/CLASS_SYMT. The analysis uses three public datasets: the Planck NPIPE (PR4) CamSpec likelihoods and lensing, available from the Planck Legacy Archive; the DESI DR2 BAO likelihoods, available from the DESI Data Release portal; and the Pantheon+ compilation, available from the Pantheon+ public release.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We thank the CLASS developers for making the Boltzmann code publicly available. The MCMC chains were produced with Cobaya. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) for editing, drafting, and supporting literature review. The theory and its derivations, the numerical implementation, the MCMC analysis, and all scientific conclusions are the work of the authors. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Appendix A. Useful exact background identities

From Eq. (30),

$$\rho_X = 3c_D \Sigma, \quad p_X = -(3c_D - 1)\Sigma, \quad \rho_X + p_X = \Sigma. \quad (\text{A.1})$$

Together with Eq. (42) this gives

$$\dot{\rho}_X + \frac{H}{c_D} \rho_X + Q = 0, \quad \dot{\Sigma} + \frac{H}{c_D} \Sigma + \frac{Q}{3c_D} = 0. \quad (\text{A.2})$$

These identities are exact within the post-transition signed-kernel effective theory at the homogeneous level.

Appendix B. Extrema of the signed kernel

Differentiating Eq. (35) gives

$$\Psi_{,r} = \frac{4(-r^2 + 4r - 1)}{(1 + r)^4}. \quad (\text{B.1})$$

The extrema satisfy $r^2 - 4r + 1 = 0$, so

$$r_{\pm} = 2 \pm \sqrt{3}. \quad (\text{B.2})$$

Evaluating the kernel at those points yields

$$\Psi(r_-) = -\frac{2\sqrt{3}}{9}, \quad \Psi(r_+) = \frac{2\sqrt{3}}{9}. \quad (\text{B.3})$$

These values are the ones used in Section 5 to derive the explicit baseline-frame crossing strip.

Appendix C. Memory-mode difference variable

Using the bare (un-stabilized) closure Eqs. (93) and (95), the perturbative mismatch variable

$$\chi \equiv \delta\rho_X - 3c_D s \quad (\text{C.1})$$

obeys

$$\chi' = \left[3c_D(1 + c_b^2) - 1 \right] \left(\nu_X + \bar{\Sigma} \frac{h'}{2} \right). \quad (\text{C.2})$$

Thus the bare closure propagates χ dynamically. In the stabilized code closure (86)–(88), χ is likewise evolved rather than zeroed, though its evolution equation differs in the super-horizon damping terms. The qualitative point is the same: the background identity $\bar{\rho}_X = 3c_D \bar{\Sigma}$ does not imply the algebraic perturbative constraint $\delta\rho_X = 3c_D \delta\Sigma$.

Appendix D. Short-wave dispersion relation

For completeness, the short-wave derivation in Section 5 is one line. In the limit $\mathcal{H} = \bar{Q} = h' = 0$, let the perturbations vary as $e^{-i\omega\eta}$. From the bare closure, Eqs. (96) give

$$\nu_X = i\omega \delta\rho_X, \quad s = (1 + c_b^2) \delta\rho_X, \quad (\text{D.1})$$

so the Euler equation becomes

$$\omega^2 \delta\rho_X = c_b^2 k^2 \delta\rho_X. \quad (\text{D.2})$$

This is the hyperbolic short-wave limit of the bare closure. In the stabilized code closure, $c_{s,X}^2$ plays the analogous role.

Appendix E. Asymptotic tail on the flat branch

When $\Psi(r) \rightarrow 0$, the sourced tail obeys

$$\dot{\rho}_X = -\frac{H}{c_D} \rho_X. \quad (\text{E.1})$$

If the tail dominates the expansion, then

$$3M_{\text{Pl}}^2 H^2 = \rho_X. \quad (\text{E.2})$$

Solving Eqs. (E.1) and (E.2) gives

$$a(t) \propto t^{2c_D}, \quad \rho_X(t) \propto t^{-2}. \quad (\text{E.3})$$

This is the asymptotic power-law behavior quoted in Section 4.5.

Appendix F. Implementation notes and validation suite

This appendix collects the practical material behind the Phase 3 chains of Section 5 and the guidance recommended for extension runs: the staged bring-up sequence, the pre-run validation checklist and its verification status, a recommended prior box for the growth-extension chains, and the sample-by-sample microscopic reconstruction recipe.

Appendix F.1. Staged bring-up sequence

The numerically sane sequence, followed for Phase 3 and recommended for any re-implementation, is:

1. validate the isolated source-free branch $\beta = 0$ in the background;
2. add the smooth transition and dark-radiation switch while keeping $\beta = 0$;
3. add the perturbation variables $\delta\rho_X$, ν_X , and s on the non-crossing branch;
4. once the scalar sector is stable, run the separate crossing branch inside (102);
5. only after the baseline implementation is reliable, consider re-deriving the source terms in the total-dark-sector Landau frame as a robustness test.

Do not turn on every structural ingredient at once. Diagnosing failures is much easier when the memory variable, the transition, and the signed loading-unloading correction are introduced one at a time.

Appendix F.2. Pre-run checks and verification status

Before any serious production MCMC run, the modified code should pass the following checks.

1. Recover the analytic source-free background scaling $\rho_X \propto a^{-1/c_D}$ when $\beta = 0$.
2. Preserve the background identity $\rho_X = 3c_D\Sigma$ to machine precision.
3. Recover Λ CDM continuously in the limit $c_D \rightarrow \infty$, $\Delta N_{\text{eff}} \rightarrow 0$, and $\beta \rightarrow 0$.
4. Verify numerically that $Q < 0$ for $r > 1$, $Q = 0$ at $r = 1$, and $Q > 0$ for $r < 1$.

5. Verify that $Q \rightarrow 0$ and $w_{\text{eff}} \rightarrow w_X$ in both asymptotic regimes $r \rightarrow \infty$ and $r \rightarrow 0$.
6. Show no gauge singularity as w_{eff} crosses -1 on the separate crossing branch.
7. Keep $\bar{\Sigma} > 0$ and $\bar{\rho}_X > 0$ for every accepted sample.
8. Enforce the explicit baseline-frame guard $4 - 3\beta\Psi(r) > 0$ over the full time window covered by the integrator.
9. Monitor $\chi = \delta\rho_X - 3c_D s$ as an evolved perturbative quantity rather than incorrectly forcing it to zero.
10. Run the full CMB+BAO+SN Ia likelihood first on the exact $\beta = 0$ branch, then on the interacting branches.
11. Show convergence of CMB and matter power spectra with respect to the switch width Δ_c and the k -sampling of the new species.

The eleven checks above are not aspirational: the background and perturbation sectors are exercised by a unit suite that mirrors the production C code line-for-line and asserts the analytic limits of Sections 4–5. On the public code (`CLASS_SYMT`, commit `b9d78d9`) the suite `test/test_afterglow_bg.py` and `test/test_afterglow_pt.py` runs 65 assertions (38 background + 27 perturbation) with 0 failures. Table F.6 maps each pre-run check to the test that establishes it and records the quantitative result. Checks 1–7 and 9 are pinned directly by the unit suite; check 8 is enforced inside the integrator and monitored as the derived chain parameter `late_branch_guard`; checks 10–11 are integration-level and were satisfied by the Phase 3 production runs themselves (three converged chains, $R - 1 \leq 6.7 \times 10^{-3}$). Check 6 combines a structural fact with a numerical transit. The structural fact is that the crossing is a property of w_{eff} alone: every $1/(1+w)$ coefficient in the perturbation system of Section 5 involves the intrinsic $1 + w_X = 1/(3c_D) > 0$, which never crosses, so no gauge singularity can arise at $w_{\text{eff}} = -1$. The numerical transit is `bg T6b`, which integrates the background at $\beta = 2.7$ — above the crossing threshold $\beta > 3\sqrt{3}/2 \approx 2.60$ of Eq. (100) and inside the Phase-3c entry gate $[2.65, 2.75]$ — and verifies that w_{eff} dips to the analytic floor $-1 + (1 - \beta\Psi_{\text{max}})/(3c_D)$ and returns smoothly, with positivity and the baseline-frame guard intact throughout; a $\beta = 2.5$ control run confirms no crossing below the threshold.

#	Check	Where verified	Result	Status
1	Source-free scaling $\rho_X \propto a^{-1/c_D}$ ($\beta = 0$)	bg T1	max rel. err 3.2×10^{-8}	✓
2	Background identity $\rho_X = 3c_D \Sigma$ preserved	bg T7–T9	ratio exact (rel. err 0)	✓
3	Continuous Λ CDM limit ($c_D \rightarrow \infty, \Delta N_{\text{eff}} \rightarrow 0, \beta \rightarrow 0$)	bg T2, pt T12	drift $6 \times 10^{-6}/6$ e-folds	✓
4	Sign of Q : < 0 ($r > 1$), 0 ($r = 1$), > 0 ($r < 1$)	bg T5	all three signs correct	✓
5	$Q \rightarrow 0$, $w_{\text{eff}} \rightarrow w_X$ as $r \rightarrow 0, \infty$	bg T3–T4, pt T15	$\Psi \rightarrow 0$ both limits; $c_{s,\text{eff}}^2 = w_X$	✓
6	No gauge singularity at $w_{\text{eff}} = -1$ crossing	bg T6b; 3c chains	crossing at $\beta = 2.7$ (3c gate): w_{eff} dips to the analytic floor $-1 + (1 - \beta \Psi_{\text{max}})/(3c_D)$ and returns smoothly; $\beta = 2.5$ control does not cross	✓
7	$\bar{\Sigma} > 0$, $\bar{\rho}_X > 0$ every sample	bg T6, T6b	positivity holds, incl. through crossing	✓
8	Guard $4 - 3\beta\Psi(r) > 0$ enforced	runtime; late_branch_guard	> 0 all accepted samples	✓
9	$\chi = \delta\rho_X - 3c_D s$ evolved, not zeroed	pt T13, T15, T17	memory mode evolves freely	✓
10	Full CMB+BAO+SNIa on $\beta = 0$ then interacting	Phase 3 chains 3a/3b/3c	converged, $R - 1 \leq 6.7 \times 10^{-3}$	✓
11	Spectra converged in Δ_c , k -sampling	production precision settings	stable to run precision	✓

Table F.6: Status of the eleven pre-run checks. “**bg** Tn” and “**pt** Tn” refer to numbered tests in `test/test_afterglow_bg.py` and `test/test_afterglow_pt.py` respectively, run on commit `b9d78d9` (65/65 assertions passing). The linear-order Bianchi residual (background) is 8.5×10^{-17} .

Appendix F.3. Recommended priors for the growth-extension chains

Phase 3 sampled the phenomenological subset defined in Table 5, holding the transition inputs $(\Delta N_{\text{eff}}, z_c, x_c)$ and the sound speed $c_{s,X}^2$ fixed. Table F.7 lists a conservative initial prior box for the growth-extension chains, which release those inputs.

Parameter	Start range	Comment
$\log_{10}(1 + z_c)$	[2, 4.5]	Keeps the transition well after BBN and usually before the observable onset of dark-energy domination.
$\log_{10} \Delta N_{\text{eff}}$	[-6, -1]	Starts from negligible dark radiation and allows a modest hidden-radiation bath.
$\log_{10} c_D$	[0.7, 2]	Corresponds to $c_D \in [5, 100]$; broad enough for theory testing while automatically satisfying $c_D > (1 + 4\beta)/3$ over the recommended β ranges.
$c_{s,X}^2$	fixed at 1/2 first; then [0.1, 1]	Hold fixed until the new perturbation sector is numerically stable.
β	[0, 2.4] or [2.65, 3.40]	Run non-crossing and crossing branches separately. The crossing branch stays inside the baseline anti-damping guard. Phase 3 sampled only the narrow entry gate [2.65, 2.75] of the crossing strip (Table 5); the growth-extension chains, with $c_{s,X}^2$ freed, are the place to open the full strip.

Table F.7: Recommended starting priors for the growth-extension chains.

Appendix F.4. Sample-by-sample microscopic reconstruction

Once a phenomenologically viable posterior exists, the microscopic parameters are reconstructed sample by sample, as applied to the 3b chain for ε_Σ in Section 5.2:

1. compute ξ from Eq. (71);

2. choose or profile over $x_c = \mathcal{O}(1)$ and compute Λ_D from Eq. (72);
3. compute $\alpha_D(M_{\text{Pl}})$ from Eq. (73);
4. compute $\rho_X(a_c^+)$ from the sampled background using Eq. (76);
5. infer ε_Σ from Eq. (74).

A posterior region that fits the data but maps to implausible microscopic values is not a success for the hidden-sector interpretation. It is only a successful late-time effective fit.

References

- [1] S. Weinberg, *The cosmological constant problem*, Rev. Mod. Phys. **61**, 1 (1989).
- [2] D. J. Gross and F. Wilczek, *Ultraviolet behavior of non-Abelian gauge theories*, Phys. Rev. Lett. **30**, 1343 (1973).
- [3] Z. Zhang, T. Xu, and Y. Chen, *Dynamical Dark Energy and the Unresolved Hubble Tension: Multi-model Constraints from DESI 2025 and Other Probes*, arXiv:2512.07281 (2025).
- [4] A. Vikman, *Can dark energy evolve to the phantom?*, Phys. Rev. D **71**, 023515 (2005).
- [5] P. Sahoo, *Stability protected phantom bound in expansion modulated cosmology*, arXiv:2601.17873 (2026).
- [6] W. Hu, *Crossing the Phantom Divide: Dark Energy Internal Degrees of Freedom*, Phys. Rev. D **71**, 047301 (2005).
- [7] S. Das, P. S. Corasaniti, and J. Khoury, *Super-acceleration as signature of dark sector interaction*, Phys. Rev. D **73**, 083509 (2006).
- [8] S. Antusch, S. F. King, and X. Wang, *Coupled dark energy and dark matter for DESI: an effective guide to the phantom divide*, arXiv:2604.08449 (2026).
- [9] A. Gómez-Valent, Z. Zheng, and L. Amendola, *Constraints on coupled dark energy in the DESI era*, arXiv:2604.12032 (2026).

- [10] M. A. Sabogal, E. Silva, R. C. Nunes, S. Kumar, and E. Di Valentino, *Sign switching in dark sector coupling interactions as a candidate for resolving cosmological tensions*, Phys. Rev. D **111**, 043531 (2025).
- [11] S. A. Adil, S. Alam, S. Sen, and J. A. Vazquez, *Dissipative cosmology and the nature of dark energy: insights from bulk viscosity with DESI DR2 observations*, arXiv:2606.05539 (2026).
- [12] P. Chanda, S. Das, and S. Das, *Dissipative dark energy can explain the DESI phantom crossing*, arXiv:2606.04886 (2026).
- [13] Y.-H. Li and X. Zhang, *Cosmic sign-reversal: non-parametric reconstruction of interacting dark energy with DESI DR2*, JCAP **12**, 018 (2025).
- [14] T. Jacobson, *Thermodynamics of spacetime: the Einstein equation of state*, Phys. Rev. Lett. **75**, 1260 (1995).
- [15] T. Padmanabhan, *Gravity: a new holographic perspective*, Int. J. Mod. Phys. D **15**, 1659 (2006).
- [16] T. Padmanabhan, *Structural aspects of gravitational dynamics and the emergent perspective of gravity*, AIP Conf. Proc. **1483**, 212 (2012).
- [17] T. Padmanabhan and H. Padmanabhan, *Cosmological constant from the emergent gravity perspective*, Int. J. Mod. Phys. D **23**, 1430011 (2014).
- [18] G. F. R. Ellis, H. van Elst, J. Murugan, and J.-P. Uzan, *On the trace-free Einstein equations as a viable alternative to general relativity*, Class. Quantum Grav. **28**, 225007 (2011).
- [19] J. E. Juknevič, D. Melnikov, and M. J. Strassler, *A Pure-Glue Hidden Valley I. States and Decays*, JHEP **07**, 055 (2009).
- [20] A. Soni and Y. Zhang, *Hidden $SU(N)$ Glueball Dark Matter*, Phys. Rev. D **93**, 115025 (2016).
- [21] L. Forestell, D. E. Morrissey, and K. Sigurdson, *Non-Abelian Dark Forces and the Relic Densities of Dark Glueballs*, Phys. Rev. D **95**, 015032 (2017).

- [22] J. Halverson, B. D. Nelson, and F. Ruehle, *String Theory and the Dark Glueball Problem*, Phys. Rev. D **95**, 043527 (2017).
- [23] W. Israel and J. M. Stewart, *Transient relativistic thermodynamics and kinetic theory*, Ann. Phys. **118**, 341 (1979).
- [24] F. S. Bemfica, M. M. Disconzi, and J. Noronha, *Causality of the Einstein-Israel-Stewart theory with bulk viscosity*, Phys. Rev. Lett. **122**, 221602 (2019).
- [25] O. F. Piattella, J. C. Fabris, and W. Zimdahl, *Bulk viscous cosmology with causal transport theory*, JCAP **05**, 029 (2011).
- [26] W. A. Hiscock and L. Lindblom, *Generic instabilities in first-order dissipative relativistic fluid theories*, Phys. Rev. D **31**, 725 (1985).
- [27] A. Poursidou, C. Skordis, and E. J. Copeland, *Models of dark matter coupled to dark energy*, Phys. Rev. D **88**, 083505 (2013).
- [28] B. Wang, E. Abdalla, F. Atrio-Barandela, and D. Pavón, *Dark matter and dark energy interactions: theoretical challenges, cosmological implications and observational signatures*, Rep. Prog. Phys. **79**, 096901 (2016).
- [29] J. Valiviita, E. Majerotto, and R. Maartens, *Instability in interacting dark energy and dark matter fluids*, JCAP **07**, 020 (2008).
- [30] A. Poursidou and T. Tram, *Reconciling CMB and structure growth measurements with dark energy interactions*, Phys. Rev. D **94**, 043518 (2016).
- [31] T. G. Martin and I. Koh, *CLASS_SYMT: the afterglow dark energy extension of CLASS*, software, https://github.com/lawdroid/CLASS_SYMT (2026).
- [32] D. Blas, J. Lesgourgues, and T. Tram, *The Cosmic Linear Anisotropy Solving System (CLASS) II: approximation schemes*, JCAP **07**, 034 (2011).
- [33] J. Torrado and A. Lewis, *Cobaya: code for Bayesian analysis of hierarchical physical models*, JCAP **05**, 057 (2021).

- [34] E. Majerotto, J. Valiviita, and R. Maartens, *Adiabatic initial conditions for perturbations in interacting dark energy models*, Mon. Not. R. Astron. Soc. **402**, 2344 (2010).
- [35] N. Yu. Astrakhantsev, V. V. Braguta, and A. Yu. Kotov, *Temperature dependence of bulk viscosity within lattice simulation of $SU(3)$ -gluodynamics*, Phys. Rev. D **98**, 054515 (2018).
- [36] G. Caldera-Cabral, R. Maartens, and B. M. Schaefer, *The growth of structure in interacting dark energy models*, JCAP **07**, 027 (2009).
- [37] A. G. Adame *et al.* (DESI Collaboration), *DESI 2024 VII: Cosmological Constraints from the Full-Shape Modeling of Clustering Measurements*, JCAP **07**, 028 (2025).
- [38] T. M. C. Abbott *et al.* (DESI Collaboration), *Dark Energy Survey Year 3 results: Cosmological constraints from galaxy clustering and weak lensing*, Phys. Rev. D **105**, 023520 (2022).
- [39] A. Heavens, Y. Fantaye, A. Mootooyaloo, H. Eggers, Z. Hosenie, S. Kroon, E. Sellentin, *Marginal likelihoods from Monte Carlo Markov chains*, arXiv:1704.03472 (2017).